

ECONOMICS

Higher 2

Syllabus 9757

Suggested Answers

Year 6 Preliminary Examination 2021



ECONOMICS

Y6 H2 Preliminary Examination 2021

Paper 9757/01
Paper 1

Case Study Question 1

- (a)(i) With reference to the data, explain whether travel for business or VFR is likely to have a higher YED value. [3]

Suggested Answer:

- VFR travel is likely to have a higher YED value.
- From Table 1, there has been an increase in income for UK and the rest of the world.
- When income increases, no. of flights to increased more significantly for VFR travel than for business travel the demand for VFR flights increases more than the increase in business flights
- Therefore, YED value for VFR travel is likely to be higher

Mark Scheme

- Identify that VFR travel has a higher YED value – 1m
- Use of relevant case material – Table 1 and Chart 1 – 1m
- Explain that VFR travel increases more than business travel for given change in income – 1m

- (a)(ii) Identify and explain one other factor that could contribute to the difference observed in Chart 1. [2]

Suggested Answer:

- One other factor could be increased globalization, specifically increased labour mobility.
- Family members could be living in a different country and households may want to visit their friends and relatives and would travel via air, hence significantly increasing the demand and quantity of VFR air travel.
- Alternatively, there could be a rise in substitutes for business travel due to improvements in technology e.g. Skype, Zoom and Teams, which could lead to a fall in demand for business travel, and hence reducing the extent of overall increase in quantity of business travel.
- Any reasonable factor explained.

Mark Scheme:

- Identification and explanation of factor that will lead to a large rise in VFR travel OR an insignificant overall rise in business travel – 2m

(b) Explain why, when faced with a fall in demand, profits tend to decrease more significantly if an industry has high fixed costs.

[2]

Suggested Answer:

- Profits = TR – TC.
- When there is a fall in demand, total revenue is likely to decrease.
- Firms with high fixed costs will have to continue incurring these fixed costs, as fixed costs cannot be varied in the short run.
- Firms with high variable costs will incur lower variable costs as variable costs decrease as DD/output decreases.
- Hence, when faced with a fall in demand, but with a relatively unchanged fixed cost, firms with a higher composition of fixed costs will face a larger decrease in profits.

Mark Scheme:

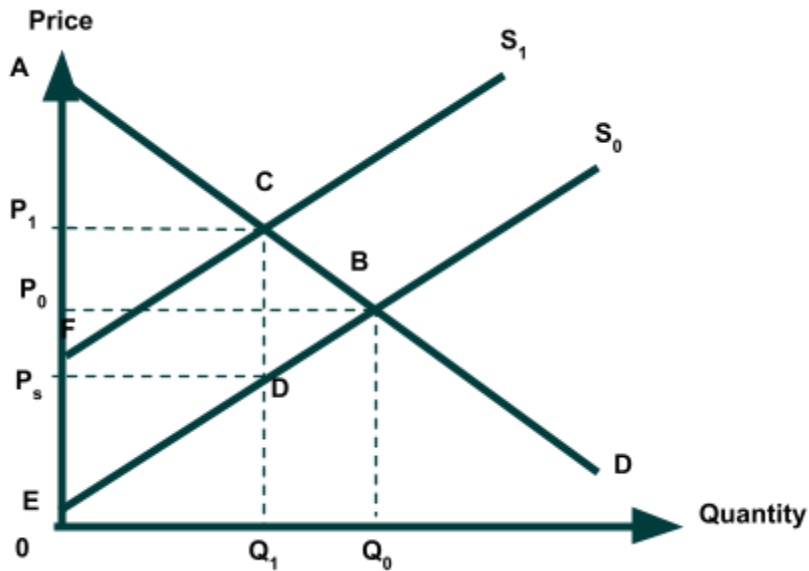
- An understanding of variable costs (varies with output or fixed costs (does not vary with output) – 1m
- Explain how a higher composition of fixed costs leads to a larger decrease in profits – 1m

(c) With the aid of a diagram, explain the impact of the imposition of eco-taxes in the market for air-travel on consumer surplus, producer surplus and government tax revenue.

[5]

Suggested Answer:

- The eco-tax is an indirect tax this is imposed on producers and acts like an increase in the cost of production (Extent depends on the type of class of tickets)
- This shifts the supply curve leftward from S0 to S1 as producers are less willing and able to produce at each price level.
- This will cause prices to increase from P0 to P1 and quantity to fall from Q0 to Q1
- Consumer surplus is defined as the difference between the price consumers are willing and able to pay and the price that they actually pay. CS decreases from ABP0 to ACP1 because the price consumers pay increases and the quantity consumed decreased.
- Producer surplus is defined as the difference between the minimum price which producers are willing and able to accept and price that they actually receive. PS decrease from P0BE to PsDE (or P1CF acceptable as the area is the same). Producer surplus decreases because the actual price producers received decreases and the quantity produced decreased.
- Government revenue = P1CDPs “tax is expected to raise 180 euros”



Mark Scheme:

- For an accurate diagram with an accompanying explanation – 2m
- For accurate identification and explanation of change in CS, PS and G revenue – 3m

(d) Discuss whether budget airlines or full-service airlines are more likely to survive.
[8]

Suggested Answer:

Introduction:

- “Survive” – make at least normal profits in the long run
- The question is an open one – important to establish the context in the arguments: normal times or during a pandemic/global recession
- Which airlines are more likely to survive is dependent on the context, firm’s strategies, and possible government intervention

Body: Compare the benefits and costs of budget airlines

Thesis: Budget Airlines are more likely to survive (in comparison to full-service airlines)

Benefits of budget airlines (over full-service airlines)

- During global recession brought about the pandemic, there could be a switch to budget airlines from full-service airlines as budget airlines are inferior to full service airlines
 - o Fall in Y – Rise in DD for budget airlines – profits rise
 - o Fall in Y – Fall in DD for full-service airlines – profits fall
- Alternatively, the fall in DD for budget airlines is lower than the fall in DD for full-service airline still able to make normal profits
 - o YED is lower for budget (normal) than full-service (luxury good)
 - o Also, budget airlines operate short-haul flights which are recovering while long haul flights run by full-service airlines have yet to recover as most borders remain close.

- o “This has a disproportionate impact on full-service airlines that still rely most on long-haul international routes.”
- Operate on a different cost structure which allows budget airlines to cover costs more easily than full-service airlines Lower LRAC
 - o e.g. less fleets to manage / storage costs of the fleets, maintenance costs, and less inventories like food, whereas full service airlines “continue to burn cash at a rate of \$5bn-\$6bn a month”.
 - o Given fall in DD/output budget airlines will face a smaller increase in AC lower fall in profits
 - o Given fall in DD/output full-service airlines will face a higher increase in AC (due to less iEOS enjoyed) greater fall in profits
- Illustrate with firm-diagram (to support their analysis) Difference in the extent of fall in DD between budget airlines and full-service airlines or to show budget airlines (rise in DD).
 - o SR: able to cover at least its TVC
 - o LR: able to cover its TC

Anti-thesis: Full-service Airlines are more likely to survive (in comparison to budget airlines)

Benefits of full service airline (over budget airlines)

- Chart 2: Airlines have past accumulated profits which can tide them over in times of recession
 - o In normal times, full service airlines operate on a much larger profit margin than budget airlines due to higher price-setting ability (less substitutes for long-haul flights) “full-service carriers rely especially heavily on lucrative business travelers”
 - o Even with a fall in demand, these firms are able to use past-supernormal profits to cover their losses.
 - o Extract 2: “Dozens of smaller airlines including FlyBe, a British low-cost carrier, Virgin Australia and Avianca, based in Latin America, have also gone bankrupt”
- Firm’s strategies that could allow them to survive during the pandemic:
 - o These strategies are undertaken by full-service national airlines rather than budget airlines: Extract 3: Meals on board stationary superjumbos, and flights to nowhere
 - o These strategies undertaken reduce the extent of fall in DD and profits allow for survival (in the short run, while waiting for borders to open)
- Full-service airlines tend to be national carriers and are more likely to be supported by their governments
 - o “Governments are used to helping airlines for reasons of national pride and to maintain international connections...”
 - o With government subsidies able to make at least normal profits/cover its losses
- Illustrate with firm-diagram (to support their analysis) Rise in DD due to firm’s strategies or fall in cost due to government subsidies

Conclusion and Overall Evaluation:

- Both types of airlines are able to survive and both can go bankrupt as well. Accept both judgements with substantiation.
- Whether it can survive depends on:
 - o The domestic market size (given that borders are closed) China vs Gulf airlines
 - o Ability of the firm to reduce costs and dampen the fall in demand FlyBe, Virgin Australia vs Ryanair, British airways vs Alitalia
 - o Extent of past accumulated profits, extent of government intervention (how important the airline is to the country – national pride or employment), ability of the airline to borrow (or raise equity or issue bonds)

- o Time period of the pandemic – if SR – both can survive, if LR – budget higher possibility of survival.

Mark Scheme:

Knowledge, Application, Understanding, Analysis		
L1	▪ One sided response – only budget or full service able to survive ▪ Lack of use of economic framework in analysis ▪ Lack of reference to case material and context of airlines. ▪ Lack of comparison between the airlines	1 - 3
L2	▪ Balanced answer ▪ Good depth of analysis with use of economic framework ▪ Use of case material and application to the context	4 - 6
Evaluation		
E1	An unexplained judgement	1
E2	Comparison of between the 2 different carriers and any insightful points	2

(e) The pandemic has caused many airlines to face severe financial difficulties.

In view of the above, evaluate the benefits and costs of government intervention in the airline industry.

[10]

Types of Government Intervention:

- Wage subsidies, tax relief or other subsidies
- Nationalisation

Thesis: Benefits

Benefits Firms / Lower Unemployment

- The government could intervene by including “wage subsidies, tax relief or other subsidies”
- Subsidies would decrease the cost of production and hence decrease costs for firms
 - o Lowering costs will lower TVC (fall in MC and AC) and ensure that $AR > AVC$ in the short run.
- In the long run, this ensures that the firm can at least make normal profits.
- Ensuring firms survive will help to save jobs that would otherwise be lost
 - o The rise in output (dampening the fall in output) will save “25 million jobs in aviation and related sectors are endangered across the world, as labour is a derived demand.
 - o This might prevent massive unemployment which impacts SOL (material and non-material welfare) adversely.
 - o This will also help to dampen the effects of the recession when more people remain employed.
- Moreover, saving these airlines are important as they are a source of “national pride” can link to non-material welfare. (Note that this cannot be the only point that you use as it lacks depth of analysis)

“Lead the recovery” – increase economic growth in the future

- “Airlines must be viable businesses so that they can lead the recovery when the pandemic is contained.”
 - Ensuring that airlines do not shut-down prevents the fall in productive capacity and rise in COP. This will prevent the dampening of recovery when AD rises again as inflationary pressure is kept in check.
 - It would also facilitate tourism and tourism expenditure (rise in X) when the pandemic is contained, and borders are open.
 - Moreover, keeping airlines maintains connectivity and ensures multinational businesses can continue to operate and that investment into the country remains feasible.
- This increase in I and X-M, which increases AD (while AS remains the same), will help to generate economic growth and lead the recovery.

Anti-thesis: Costs

Opportunity cost

- Excessive subsidies might be very costly for the government.
 - “Dutch €10 billion ; Germany €9 billion; Sweden €455 million (loan)”
 - This might create a budget deficit for many governments, which may create the need to incur significant public sector debt. In the longer run, this could dampen economic growth as debt has to be financed through fall in G or rise in tax rates.
- There would also be severe opportunity costs as funds could be used to finance healthcare spending in light of the pandemic, where funds for vaccines, quarantine facilities and test kits should be prioritized lower non-material welfare

X-inefficiency

- Excessive subsidies or nationalization might create also lead to X-inefficiency. For the latter, this is because government owned entities might not possess the profit incentive.
 - Airlines would have less incentive to cut cost
 - Airlines will produce at a point above the LRAC
 - Hence, nationalisation might be worsening productive inefficiency

Conclusion and Overall Evaluation:

- In the context of the pandemic, the benefits if government intervention is likely to outweigh the costs as:
 - Once-off severe pandemic significant costs on the economy so the benefits in preventing the shutting down of airlines can be significant.
 - Costs of intervention can be reduced. Governments can refine conditions of the subsidies/bailout to reduce X-inefficiency e.g. subsidies can be given on the condition that airline also restructure and cut costs.
- The extent of the benefits depends on how dependence the country is to the airline industry in the context of growth and employment.
- The extent of costs depends on the government's budget and its reserves, and the type of government intervention.

Mark Scheme:

Knowledge, Application, Understanding, Analysis
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L1	▪ Conceptual errors ▪ Lack use of economic framework in explaining the benefits and costs ▪ Limited use of case evidence	1 - 4
L2	▪ Sufficient scope – 3 points in total ▪ Balanced answer – both benefits and costs explained ▪ Use of economic framework ▪ Good application of case evidence	5 - 7
Evaluation		
E1	An unexplained judgement Weak justification on the extent of costs and benefits	1
E2	Insightful points made on the extent of costs and benefits and cost vs benefits	2 - 3

Case Study Question 2

- (a) (i) Describe the trend of crude oil prices between 2012 and 2020.
[2]

Suggested Answer

- Oil price fell between 2012 to 2020.
- Refinement: There is an increase between 2016 to 2018 **OR** There is a sharp fall between 2014 to 2015

Mark Scheme:

- General trend – 1m
- Refinement – 1m

- (ii) Explain how the balance of trade of the Gulf states may be affected by the above trend.
[2]

Suggested Answer

- Balance of trade: Value of Exports ($P_x.Q_x$) – Value of Imports ($P_m.Q_m$)
- Crude oil is an essential factor of production used by all industries worldwide (fuel, electricity etc) and it has few substitutes demand for oil is price-inelastic.
- Given that $PED < 1$, a fall in price of oil will lead to a less than proportionate rise in quantity demanded. Thus, the export revenue ($P_x.Q_x$) of the oil exporting Gulf states will likely fall, leading to a negative impact on the BOT, ceteris paribus.

Mark Scheme:

- Identify the PED value of oil – 1m
- Explain the link between price of oil and value of oil exports – 1m

- (b) Explain the difference in the total labour force participation rate between Saudi Arabia and Singapore.
[3]

Suggested Answer

- The total labour force participation rate in Singapore (SG) is higher than that in Saudi Arabia.
- Identify and explain any of the following reason:
 - o A relatively small proportion of 22% of the women in Saudi Arabia work, versus Singapore's 62%. This could be due to social or legal or cultural norms that do not encourage women in Saudi Arabia to take up work outside the home. E.g. strict laws do not allow women to work within the same premises making it difficult for employers to employ women.

OR

- o The Gulf states' welfare for its citizens are so generous that the opportunity cost of not having a job is relatively low in SA as compared to SG and so Saudi Arabians, particularly the women, may chose not to work. The state provides free or heavy subsidies for healthcare, education and utilities for its people and they are able to enjoy a good standard of living (non-material) even without employment.

Mark Scheme:

- Correct identification of the difference in labour participation rate – 1m
- Any well-explained reason – 2m

(c) **With reference to Table 3, what does the data suggest about the cost of living in Saudi Arabia as compared to that in Singapore. Justify your answer.**

[2]**Suggested Answer**

- Cost of living is higher in Singapore (SG) than Saudi Arabia.
- GDP per capita (PPP USD) eliminates differences in cost of living between SG and Saudi Arabia and equalises the purchasing power of different currencies by eliminating the difference in price levels.
- From Table 2, Singapore's GDP / capita (constant USD) and GDP/capita (PPP USD) are both higher than Saudi Arabia's. Singapore's GDP / capita (constant USD) is 2.9 times that of Saudi Arabia, which suggest that the average citizen in SG can purchase and consume 2.9 times more than an average citizen in Saudi Arabia for a common basket of goods and services. However, when we use GDP / capita (PPP USD) to eliminate the price level difference in the 2 countries, Singapore's GDP / capita (PPP USD) is only 2.1 times that of Saudi Arabia, which suggest that the average citizen in SG can purchase and consume a lower 2.1 times more than an average citizen in Saudi Arabia for the same basket of goods and services. Thus, the cost of living in SG is higher than in Saudi Arabia.

Mark Scheme:

- Correct identification of country – 1m
- Accurate justification – 1m

(d) **With reference to Extract 6 and your own knowledge, explain how you would expect the size of the multiplier to differ between Saudi Arabia and Singapore.** **[3]**

Suggested Answer

- The multiplier is a coefficient that shows the multiple increase in the national income for a given increase in spending and the multiplier size is calculated by the formula: $k = 1 / (MPS + MPT + MPM)$. The multiplier size is determined by the size of leakages, namely, taxes, savings or import expenditure in each round of the multiplier process. The larger the MPW, the larger will be the leakage for given increase in NY, and thus a smaller overall increase in NY small size of k.

- Saudi Arabia's size of k is likely to be larger than Singapore's because of a smaller MPS and MPT (or smaller leakages due to lower savings and tax payment).
- MPS - The leakage in terms of savings for a one dollar increase in income is likely to be smaller in Saudi Arabia because of the generous state welfare in healthcare, education and utilities (Extract 6) do not incentivise savings for precautionary reasons or for retirement. Comparatively, the lower level of welfare in Singapore is likely to encourage savings for precautionary reasons. In Singapore, there is also the monthly savings for retirement needs through the compulsory contribution to one's CPF account.
- MPT - Unlike Singapore, personal income tax is not payable in Saudi Arabia (Extract 6: no individual income tax scheme) and so there is no leakage in the circular flow of income. MPT in Singapore is higher because personal income tax is payable.

Mark Scheme:

- *Determination of size of k – 1m*
- *Comparison of 2 differences in the size of leakages – 2m*
- *Lack comparison – max 2m*

(e) **Discuss whether Singapore has a higher standard of living than Saudi Arabia.**
[10]

Suggested Answer

Introduction

- SOL of a country refers to the level of economic welfare of an average person and it includes the material and the non-material welfare.
Material welfare refers to the quantity of goods and services consumed by the average person in a country in a given time period.
____Non-material welfare refers to the qualitative aspects of life.

Thesis: Singapore seems to have a higher SOL than Saudi Arabia

- From Table 3 – Real GDP per capita (PPP USD) is more than 2 times higher in S'pore, meaning the average person in S'pore has greater ability to buy more goods and services than an average resident in Saudi Arabia. Higher material welfare for an average resident in SG.
- Singapore's unemployment rate is lower than Saudi Arabia. Higher percentage of workers in the labour force employed. Increased income and purchasing power of citizens to consume goods and services. Material welfare of Singapore is higher relative to Saudi Arabia. Further, since greater proportion of labour force employed in SG, greater access to healthcare and education services to increase quality of life or citizens enjoy a lower crime rate and less social unrest. Non-material welfare of Singapore is higher relative to Saudi Arabia.
- In addition, non-material welfare is lower for Saudi Arabia given social and legal restriction on women in public and work place. Not having the ease or freedom to move about physically in the country or to pursue one's aspirations is an indication of a lower quality of life.
- The ease of doing business is low in Saudi Arabia, ranked 92nd in the world and a relatively high level of corruption is likely to extend to other areas in society where money can be used to oil transactions and this is likely beneficial to the rich where they can enjoy privileges through bribery. It indicates the lack of inclusiveness or equity where benefits may not be distributed evenly.

Anti-thesis: Saudi Arabia may have a higher SOL than SG

- Higher labour force participation in SG than Saudi Arabia implies relatively larger proportion of adult Singaporeans hold a job and so the average person enjoys a lower number of leisure hours and may suffer from more work-related stress non-material SOL lower for SG than Saudi Arabia.
- No income tax in Saudi Arabia payable implies the average person may have a higher disposable income and purchasing power to consume goods and services in Saudi Arabia despite lower GDP/capita. While in Singapore, though GDP/capita is higher, but disposable income is reduced by a progressive income tax structure.
- Presence of non-marketed items may not be factored in overall GDP statistics for Saudi Arabia. Saudi Arabia may have more non-marketed activities due to the lower female participation rate in the labour force compared to SG. A greater proportion of women in SA are home-makers who stay at home to provide productive activities that are not captured in the calculation of GDP. However, similar service of housework would be done in SG by paid cleaners and recorded as a transaction in the market for home-cleaning services in SG unpaid household chores which are not recorded may underestimate GDP statistics and the SOL in Saudi Arabia. SOL higher for SA than SG.
- Given lower female participation rate in SA, the average family may enjoy a better quality of life given that a larger proportion of the women stay home and so spend more time with family and care for children.
- The Saudi is very rich in oil (30% of GDP), is the biggest oil exporter and oil accounts for 80% of its government revenue. This allows for the provision of healthcare, which seems to be heavily subsidised if not free and that is likely to indicate a higher quality of life when one can easily seek treatment for illnesses. This is also likely to indicate a higher life expectancy and a lower infant mortality rate.

OR

- Saudi Arabia may have a higher material welfare due to generous state benefits provided for citizens. Saudi Arabia gives citizens subsidised fuel, electricity and water and loans and grants, wedding goods/services and overseas education etc greater consumption of goods and services relative to SG higher SOL for Saudi Arabia than SG.

OR

- Stress level generally may be somewhat lower in Saudi Arabia because many basic goods and services for the households are provided or subsidised by the state (fuel, electricity and water, loans/grants, wedding goods/services and overseas education etc). The opportunity cost of unemployment and the related stress is therefore relatively lower compared to Singapore where the welfare provision is likely to be lower.

Conclusion and Overall Evaluation:

- Difficult to conclude whether Singapore has a higher SOL than Saudi Arabia.
- The higher GDP/ capita (PPP) of SG may hint towards higher material welfare in SG than in Saudi Arabia though additional indicators like the Gini coefficient may improve the comparison. Where the qualitative aspect of SOL is concerned, it is difficult to judge whether SG or Saudi Arabia has a higher level. This is because it is very difficult to put a monetary value to the indicators of quality of life, for example, stress level, leisure hours, family time etc. And this is made even more difficult because the 2 countries have very different values and many judgments on quality of life depend on the culture and social norms in SG and Saudi Arabia, which in this case are vastly different.

Mark Scheme:

Knowledge, Application, Understanding, Analysis		
L1	- Shows weak knowledge of SOL (material and non-material). - No comparison, one sided answer. Explanation is limited to one country. - Glaring errors evident. - No use of data or contextual evidence.	1 - 3
L2	- Able to discuss and compare using the contextual evidence and SOL yardsticks/indicators given in the case study. - Answer covers sufficient scope in terms of material and non-material SOL. - Answer has sufficient depth and well-developed economic analysis that demonstrated strong understanding of material and non-material SOL indicators used.	4 - 6
Evaluation		
E1	An unexplained judgement An unexplained evaluative conclusion	1
E2	Evaluative assessment supported by economic analysis	2

- (f) Discuss the view that inclusive and sustainable economic growth can be achieved through “fostering innovation” in the Singapore economy. [8]

Suggested Answer**Introduction**

- Inclusive growth refers to a rate of growth that is sustained over a period of time, is broad-based across economic sectors, and creates productive employment opportunities for the majority of the country's population.
- Sustainable growth indicates a rate of growth that can be maintained over time without creating other significant economic problems (such as depleted resources and environmental problems), particularly for future generations.
- In the pursuit of sustainable or inclusive growth, sustained growth needs to be first attained. Sustained economic growth refers to a positive and stable, non-inflationary growth rate over an extended period of time. It can occur when the AD and AS increase in tandem, causing an increase in real output without significant upward pressure on the general price level.

Thesis 1: Fostering Innovation helps to achieve inclusive EG

- Investment in R&D and the adoption of technology (Extract 7: Research, Innovation and Enterprise 2020 plan and Extract 8: invest in research into energy and resource efficiency technologies) process innovation from better techniques of production increase in productivity and efficiency of production processes reduce unit cost of production and increase the productive capacity of the economy as resources are more productive, thus increasing potential output, shifting the AS curve outwards.

This measure is especially helpful to Singapore given a manpower lean economy (Ext 7)

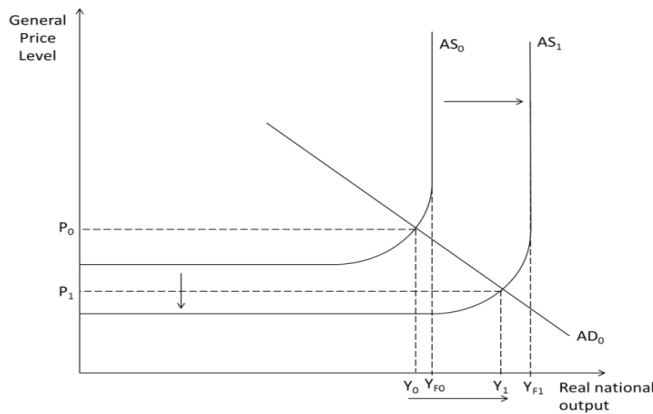


Figure 1

- Fostering Innovation through the use of technology and automation to complement labour (Ext 8: aims to create better jobs for workers and provide stronger support; Ext 7: workers to adapt to this changing nature of work) helps to increase labour productivity through more efficient capital-labour ratio increasing demand for lower skilled labour and increase wages of lower skilled workers when they move to higher value added industries as firms are more willing to hire more productive local workers with a better set of skills narrow income (wage) gap between high skilled and low skilled workers achieve inclusive growth. This is especially so for lower skilled workers and older workers as inferred in Extract 7.
- Increase in RNY increase in tax revenue and increased ability to redistribute income to lower income groups. Eg. Transfer payments or subsidies to lower income households increase in disposable income and increases in purchasing power of lower income group and help narrow income gap

Thesis 2: Fostering Innovation helps to achieve sustainable EG

- Innovation reduce the use and demand of scarce resources / limited resources resources through process innovation or through the use of cleaner technology (Ext 8: govt. investment in research in energy and resource efficiency technologies)

increased efficiency / productivity / quality of scarce resources or production processes increase in AS (AS curve shifts out) as both actual and potential EG increases sustained EG and allow for further increase in AD without inflationary pressure sustainability of growth given reduced strain on resource and an increase in potential output (Ext 7: find new ways of overcoming our constraints).

As firms switch to cleaner inputs or technology reduce the level of pollution and carbon footprints (Ext 8: environment sustainability in terms of resilience in low carbon and resources constraint future) reducing the strain on resources achieve a more sustainable EG.

Anti-thesis 1: Limitations of innovation in achieving inclusive EG

- Productivity gains may hinder the attainment of inclusive growth e.g. innovations tied to investment in technology or increased usage of capital inputs e.g. Robotics which may displace low-skilled workers increase in unemployment and fall in wages of low skilled workers increase income gap/ disparity between low and high skilled workers.
- Innovation and the corresponding productivity gains which lead to higher economic growth and jobs creation may not be distributed equally across all sectors e.g. Labour intensive sector where

productivity gains from innovation slower to be realized relative to capital intensive / tech intensive sectors e.g. Ext 7: new start up and venture capital industry inclusive growth not be achieved and may even result in a widening of the income gap and a fall in material and non-material SOL.

- Effectiveness of innovation to complement labour largely depends on the mindset as well as receptivity of training workers. This is especially true for lower-skilled and older workers where there is often a steep learning curve to acquire new knowledge and skills. This unreceptiveness of workers will limit the intended effects of higher labour productivity and inclusive growth (Ext 7: not an easy task for older workers in their 50s and 60s).
- Efforts to innovate may be difficult given a narrower technology gap now in SG (Ext.7), Singapore has to expand its own capacity to innovate which may take time bear fruit and ability to attract FDI may be limited / slow. As innovation is long term in nature as it takes time to achieve results, the measure may not be as effective in achieving inclusive growth in the short run.

Anti-thesis 2: Limitations of innovation in achieving sustainable EG

- Innovation via R&D development in energy and resource efficiency tech may not guarantee results and there is a long time lag. For R&D, small economies like Singapore face resource constraints (Ext 8), and thus face difficulty in terms of the type of alternative energy sources that can be used / developed
 - limited land size lack of ability (eg. space constraint) to use alternative sources of clean energy across all sectors, eg, solar or wind energy to reduce carbon emission.
- Fostering innovation and prioritising technological pursuit might also end up damaging the economy if it results in pollution. Depending on the type of technology pursued, there might be environmental damage.
 - In Singapore, accelerated use of digitalization (Ext 8) increasing use of digital devices increased use of electricity
- Digitalisation might also result in pollution from the production of IT hardware or from e-waste i.e. used electrical and electronic equipment. conflict with the goal of achieving sustainable economic growth.

Conclusion and Overall Evaluation:

- Given the small and open nature of SG economy, investing in sustainability of resources as well as in manpower via innovative practices is important as it helps boost resilience of SG economy to external shock and helps achieve a more inclusive and sustainable growth.
- However, innovation efforts to increase inclusive and sustainable growth are focused largely on improving supply-side conditions. Without an active and continual growth in aggregate demand and with SG subject to the vagaries of the global economy (Ext 7), economic growth may not be easily sustained, and be neither sustainable nor inclusive. Also, using innovation to achieve both inclusive and sustainable growth is a long term measure, and is ineffective in the shorter run.
- Need for multi-pronged approach: Given Singapore's small domestic and labour market, it is essential that fostering innovation is coupled with interventionist measure of education and retraining to ensure sustainable and inclusive growth. The SG govt should also continue with a range of demand management policies to ensure the expansion of aggregate demand, as well as other policies to curb environmental degradation and improve income distribution in the country.

Mark Scheme:

<i>Knowledge, Application, Understanding, Analysis</i>		
<i>L1</i>	▪ <i>Smattering of points not directly linked to question</i> ▪ <i>No theoretical framework and/or glaring conceptual errors</i>	<i>1 - 4</i>

	▪ One-sided weak analysis with gaps in explaining how innovation is effective in achieving both inclusive and sustainable economic growth. ▪ No reference to case material.	
L2	• Balanced argument and good scope that includes explanation of how fostering innovation is effective as well as limited in achieving both inclusive and sustainable economic growth. • A well explained economic analysis with reference to accurate and well-labelled diagram/s. ▪ Effective use of case evidence to develop and support analysis.	5 - 7
Evaluation		
E1	An unexplained judgement An unexplained evaluative conclusion/comment	1
E2	Evaluative assessment supported by economic analysis	2 - 3

Paper 9757/02
Paper 2

Question 1

The world demand for semiconductors has surged far beyond the capacity for supply. Outbreaks of Covid-19 cases created delays in the global supply chain. This is crippling players in industries such as automobiles and smartphones.

Explain the demand and supply factors that determine the price of semiconductors and discuss how consumers and producers in different product markets may be impacted by this price change.

[25]

Suggested Answer:

Introduction:

- Semiconductors are used as factors of production in many products such as automobiles, smartphones, washing machines, computers and etc. They are essential input in every electronic product.
- Price of semiconductors is affected by demand and supply of semiconductors.

Body

1. Demand factors (Derived Demand)

- Increase in income or affluence of households globally: As income increases globally, e.g., increase in middle income groups in China and India, this drives up demand for computers, washing machines, smartphones. Therefore, this increases the demand for semiconductors. Especially when the demand for these goods is income-elastic, demand increases more than proportionately when income increases. Hence, this will result in a significant increase in demand for semiconductors.

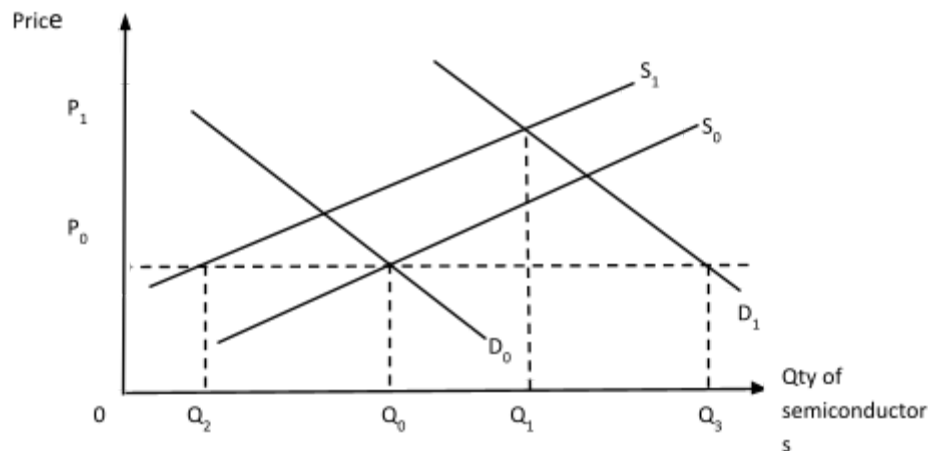
- Changes in working/schooling arrangements during pandemic: Since the beginning of this pandemic in 2020, globally, working adults are encouraged to work from home and students have home based learning. Demand for computers, laptops, webcams are needed to carry out such activities. Such sudden increase in demand for these devices therefore drives up the demand for semiconductors.
- Rise in 5G technology: Many improved products work on 5G technology because they provide enhanced speed, higher quality video streaming and efficiency. From the basic product like smartphones to sophisticated product like smart vehicles, remote robotic surgery and automation in every industry, 5G technology improves the functions of such devices. Improved semiconductors are required to power such devices. This increases the demand for semiconductors.
- These factors that increase the demand for semiconductors is illustrated by the rightward shift of the demand curve.

2. Supply factors

- Closure of factories/ borders during pandemic: During the early stage of the pandemic, when economies were locked down. Semiconductor factories were closed and production came to a halt. This decreases the supply of semiconductors.
- During pandemic, when borders are closed, this limited the mobility of labour. Such fall in the supply of labour may have increased the cost of production which disincentivise the supply of semiconductors.
- This is illustrated by a leftward shift of the supply curve of semiconductors.
- The PES of semiconductors may be highly price inelastic since 'far beyond the capacity for supply'. This will result in a more than proportionate increase in price when demand increases.

3. Price determination of semiconductors

- When 'world demand for semiconductors has surged far beyond the capacity for supply', this means that increase in demand is greater than the decrease in supply of semiconductors, as reflected by the larger rightward shift in demand curve from D_0 to D_1 than the leftward shift in supply curve from S_0 to S_1 .



At the initial price OP_0 , a shortage Q_2Q_3 will occur. There will be an upward pressure on price as consumers try to outbid one another for the existing supplies.

- As the price rises, it becomes more profitable for firms to increase their output. Quantity supplied increases as reflected by an upward movement along the new supply curve S_1 . The higher price

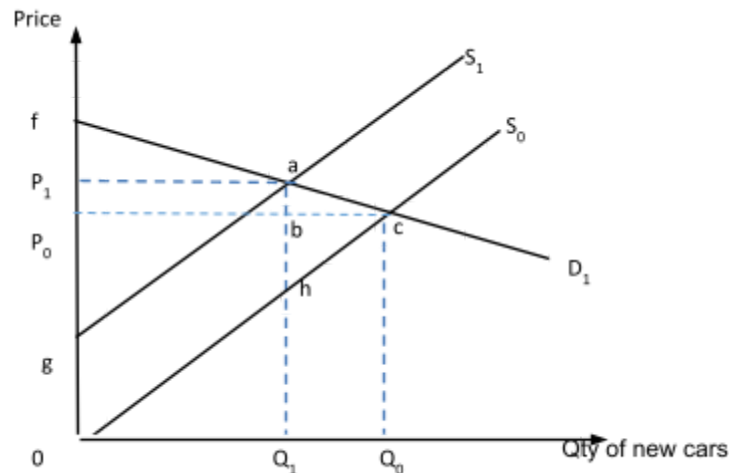
also reduces consumers' ability to pay and willingness which leads to fall in quantity demanded as reflected by an upward movement along the new demand curve D_1 .

- The process continues until the shortage is eliminated. At the new equilibrium, price has risen to OP_1 and quantity increased to OQ_1 .

4. Explain how product markets may be impacted by the rise in price of semiconductor, ceteris paribus.

(a) Automobile market (new cars)

- Since semiconductors are used as input of automobiles, eg, the display board in the car, cell phone integration in the car. As price of semiconductors increase, this will increase the cost of production of automobiles. This reduces the profitability of car producers. Firms will be only willing to supply fewer cars at each price. The supply of cars will fall and the supply curve shifts leftwards from S_0 to S_1 . This will result in an increase in the price of new cars.
- As the demand for new cars is price elastic since the price of new car is a large proportion of the buyers' income or there are substitutes to private transport, when price of new cars increase from P_0 to P_1 , this will result in a more than proportionate decrease in the quantity demanded for new cars from Q_0 to Q_1 .

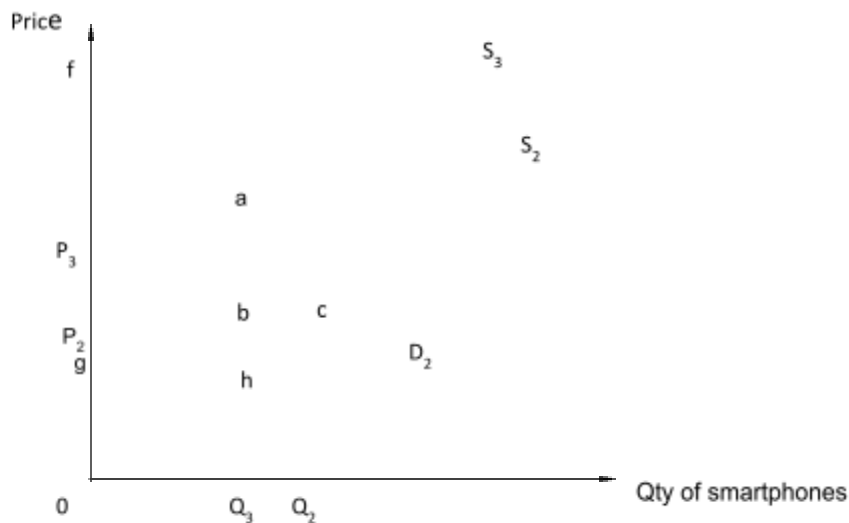


- Impact on Consumers:
 - o Higher price for a lower quantity resulting a fall in expenditure on new cars. This is because the gain in revenue due to the rise in price (Area P_0P_1ab) is less than the loss in revenue resulting from the decrease in quantity demanded (Area Q_0Q_1bc). The final revenue (Area OP_1aQ_1) is smaller than initial revenue (Area OP_0cQ_0). Hence, a rise in price of new cars will lead to an overall fall in total expenditure of the consumers.
 - o Fall in consumer surplus from P_0fc to P_1fa . Since the quantity of goods consumed decreases and the price paid is at a higher level, consumers surplus has definitely fallen, and consumer welfare worsened.
 - o The proportion of the increase in cost of production of new cars that is borne by consumers is distance ab .
- Impact on producers:

- o Higher price but lower quantity supplied resulting a fall in revenue, as revenue received must equal to expenditure of the consumers.
- o Producer surplus decrease from OP_0c to gP_1a . As producers sell lower quantity of goods while incur a higher costs of production, the producer surplus has definitely worsen.
- o The proportion of the increase in cost of production of new cars is borne more by the producers, as demand for new cars is more price elastic than the supply of new cars. This is illustrated by the distance bh .
- Other acceptable arguments:
 - o In certain developed or developing countries, price of cars may not be as high as in Singapore, therefore the demand will be more price inelastic. The discussion of impact on consumers and producers will be the same as (b).
 - o Possible extension and evaluation: different type of cars will be affected differently electric cars definitely make use of more semiconductors than traditional cars. The increase in cost of production will therefore be relatively higher resulting in much higher price of electric cars than traditional cars. Hence, the demand will be more price elastic than the traditional cars. The impact on consumers of electric cars will be much worse off.
 - o Coupled with an increase in demand based on income/changes in taste and preferences, the overall impact could also be different.

(b) Smartphones market

- The proportion of the cost of semiconductor out of the cost of smartphone maybe much greater than the proportion of cost of semiconductor out of cost of new car, hence, the increase in cost of production of semiconductor may be higher than the increase in cost of production of new cars. Then the leftward shift of supply curve of smartphone will be much more than the ss curve of new car.
- The demand for smartphone however can be more price inelastic than automobiles, this is because the proportion of income is much smaller or lack of substitutes. The increase in price of smartphone will cause a less than proportionate decrease in quantity demanded.



- Impact on consumers:

- o Higher price and lower quantity demanded. There will be an increase in expenditure, since area P_3abP_2 is greater than bcQ_2Q_3 , this will result in an overall increase in consumer expenditure.
- o A fall in consumer surplus from cP_2f to aP_3f .
- o Consumers end up bearing a greater proportion of the increase in cost of production of smartphones since the demand is relatively more price inelastic than the supply. This is represented by distance ab .
- Impact on producers:
 - o Higher price and lower quantity supplied resulting in higher revenue.
 - o Fall in producer surplus from OP_2C to P_3ag .
 - o Producers share of the increase in cost of production is only distance bh .
- Possible EV: different type of digital consumer products (laptops, smartphones, digital cameras, washing machines...) will be affected differently based on differences in extent of increase in COP and PED/PES. Coupled with an increase in demand based on income/changes in taste and preferences, the overall impact could also be different.

(c) Any other acceptable markets

Conclusion and Overall Evaluation

- Both demand and supply factors have resulted in a significant increase in the price of semiconductors. Research has been carried out to find substitute of semiconductors but is still on going, and this could change in the longer run.
- The markets for products that need semiconductors as the factor input will be adversely impacted. The more semiconductors the product needs, the higher the hike in price of the product. How much of the price hike in the product is borne by the consumers and producers will however depend on the price elasticity of demand relative to the supply of the product.
- Consumers and producers are definitely worse off for now. If consumers can hold their demand for a longer time or replace with a second-hand product such as second hand cars, they will not have to bear the brunt of the rise in price of semiconductors. Producers are definitely much worse off, if they hold their production because they can't get hold of semiconductors, their revenue will definitely decrease. But if they continue to produce with the pricier semiconductor, they may be worse off too if the demand for their product is price elastic.
- In the long run, when countries' effort to set up more factories to produce semiconductors or firms carry out research and development to substitute the use of semiconductors, the price of the semiconductors is likely to decrease.
- The ceteris paribus assumptions may not be true. In reality, many changes may happen at the same time, hence the outcome discussed in the different markets may therefore differ.

Mark Scheme:

Knowledge, Application, Understanding, Analysis		
L1	• Little / non-existent use of economic theory / framework in the explanation. • An answer that does not show that meaning of question is understood • An answer that give only provide dd or ss factors to explain increase in price of semiconductors.	1 – 8

L2	• Answer is relevant but explanation may be incompletely explained. • An answer that has some gaps in explanation and/or some conceptual errors. • Factors discussed (both dd and ss covered), PAP done. • Cover 1 product market in detail for both consumers and producers.	9 - 14
L3	• An answer that has well defined and selected factors, containing sufficient scope of points discussed. • Factors mentioned are well-developed with rigorous and effective use of economic framework.	15 - 20
Evaluation		
E1	An unexplained judgement/conclusion/comment	1
E2	Evaluative assessment supported by economic analysis Some substantiation of an evaluative comment and/or conclusion	2 - 3
E3	In depth or well explained insights.	4 - 5

Question 2

President Biden has signed a new executive order to promote competition in various industries, with the aim of reducing monopoly power of large firms. The order calls for regulators to enact measures including increasing their scrutiny of mergers and preventing acquisitions of smaller brands.

Source: www.whitehouse.gov

(a) Explain how an increase in the level of competition would likely affect consumers, firms and the American economy.

[10]

(b) Discuss the strategies that firms can adopt in response to the rise in the level of competition.

[15]

Suggested Answer

Part (a)

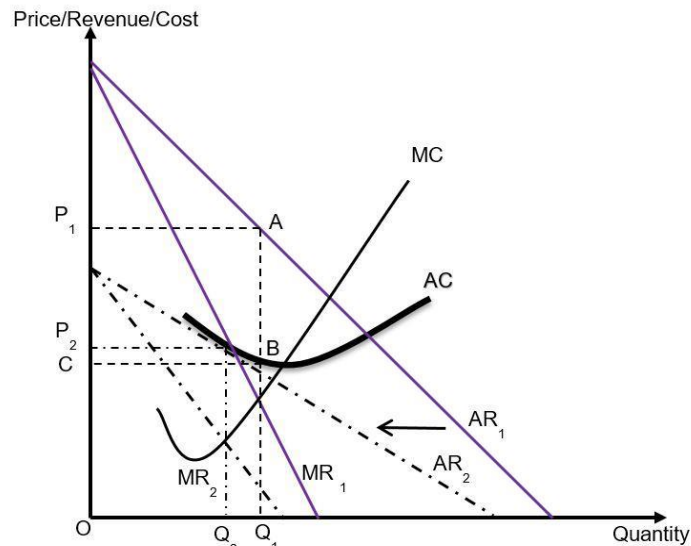
Introduction

- **Clarify what is meant by the rise in the level of competition** – can be measured by the number of firms in the industry.
- **Clarify what is meant by the “likely effect”** – desirable and undesirable impact on consumers, producers and the American economy.
- **The impact can be measured using the following indicators** – price, output, choice/variety, quality of products, profits, macroeconomic objectives

Explain how government intervention can reduce monopoly power of large firms and promote competition.

- Government intervention in the form of anti-trust laws will lower the barriers to entry in many of the industries such as telecommunication, technology industry, pharmaceutical, banking industry that face high statutory and strategic BTE.

- With lower BTE, the entry of new firms would lead to increase in the number of firms in the industry and reduce the market share of the incumbent firm, assuming the market size remains constant. For example, the entry of Apple Inc., Alphabet Inc.(Google), YouTube, TikTok, Snap, Twitter and Amazon has increased the level of competition in the market for advertising where that compete with Facebook that sell online advertising to marketers, as well as other companies that provide platforms for communicating and sharing content among users' various social networks.
- Hence, this leads to a fall in the demand for Facebook's online advertising platform and increase in price elasticity of demand for the service provided as illustrated in diagram below.
- As demand falls and becomes more price elastic due to the increase in the number of substitutes, AR and MR curves shift from AR_1 to AR_2 and MR_1 to MR_2 respectively, and become gentler. The profit-maximising equilibrium price and output fall from P_1, Q_1 to P_2, Q_2 .



Explain the likely impact on consumers.

- *Desirable impact:*
 - With a possible increase in the market supply, consumer surplus rises as they get to enjoy a larger quantity of goods and services. Moreover, welfare also rises as they pay a lower price of P_1 . With lower prices, lower income American households will also enjoy these goods and services that were previously inaccessible. This will reduce inequity.
 - Incumbent firms such as AT&T and Bank of America will be forced to improve the quality of their services in the face of potential loss of market share contesting in the market. As such, consumers can now enjoy better quality banking and telco services, raising the consumer welfare.
 - With potential new entrants, consumers will have a larger number of firms to choose from that provide similar product or services. This will improve choice and hence consumer welfare. For example, with the introduction of new telcos, this will help to increase product variety resulting in more choices of mobile data services for consumers such as higher data caps and free or discounted bundled services like music service Spotify.
 - In addition, the intensification of competition incentivises firms to invest in research and development. In the long run, the results of product and process innovation may benefit consumers. Consumers get to benefit from the resultant increase in range of products and a further fall in price as AC and MC fall – a result of better production methods.

- *Undesirable impact:*

- But it can also be argued that with more competition, the decrease in supernormal profits reduces the ability of firms to invest in R&D. In this regard, consumers' welfare may fall instead.
- And it remains to be seen if smaller entrants can survive in the industry as incumbents launch pricing and non-pricing strategies in the face of competition. These strategies require the use of massive funds and past supernormal profits. If these rivals are edged out, the market will continue to be dominated by the incumbents in the long-run, leaving them with high degree of market power. This will result in consumers continuing to pay a high price and lower consumer surplus in the long-run.

Explain the likely impact on producers.

- With increased competition, the incumbent firm would experience a decrease in demand for its products from AR_1 to AR_2 , ceteris paribus. The profit-maximising output is now lower at Q_2 , with the firm charging a lower price of P_2 . The firm's initial supernormal profits of area P_1CAB falls, and can even fall to normal profits where $AR=AC$ as shown above.
- As for the entrants, there is unlikely to be any more space to accommodate more firms in certain industries. For example, in the smartphone market, the saturation is close to 100% and hence, the likelihood of a new entrant surviving the rise in the level of competition is very low.
- The fall in output reduces the internal economies of scale of the firm. In this case, the firm may be operating further away from the minimum efficient scale, and is thus more productively inefficient from society's perspective,
- With the rise in competition, incumbents may become less X-inefficient in a bid to retain their supernormal profits. As such, they will likely reduce wastage in the form of lavish and unnecessary spending, and thus produce closer to a point on the LRAC curve. Thus, they are less productively inefficient from their own perspective.

Explain the likely impact on the American economy.

- As competition intensifies, firms are incentivised to be X-efficient and dynamic efficient. As they invest in process innovation, the better production methods can lead to a fall in MC and AC. For the economy as a whole, the lower cost of production leads to an increase in the aggregate supply of goods and services. The lower general price level encourages consumption and also makes the exports become more price competitive relative to other countries. This increases the quantity demanded of exports and assuming the $PED_x > 1$, there will be an increase in export revenue. At the same time, Americans will switch away from imported goods and services to the relatively cheaper domestically produced goods and services. This reduces the demand for imports. As such, there will be an improvement in national income and hence balance of trade.
- The resultant increase in range and quality of products due to product innovation can also be a boost to C and X-M.

Conclusion

- The impact of higher level of competition is likely to be desirable on consumers, producers and the American economy in the long-run. However, the overall impact depends on the industry and the degree of competition faced by firms in these industries as a result of Biden's anti-trust revolution.

Mark Scheme:

Knowledge, Application, Understanding, Analysis
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L1	• Little / non-existent use of economic theory / answer that is largely descriptive knowledge with little or no economic terminology applied. • An answer that does not show that meaning of question is understood	1 – 4
L2	• Answer did not explain the impact on either - o consumers - o firms - o or macroeconomic performance of the US. • Some gaps in explanation (depth lacking in some areas / some conceptual errors)	5 - 7
L3	• Answer explains the impact on consumers, firms and macroeconomic performance of the US. • Answer has sufficient depth and economic terminology is well used to demonstrate rigour	8 - 10

Part (b)

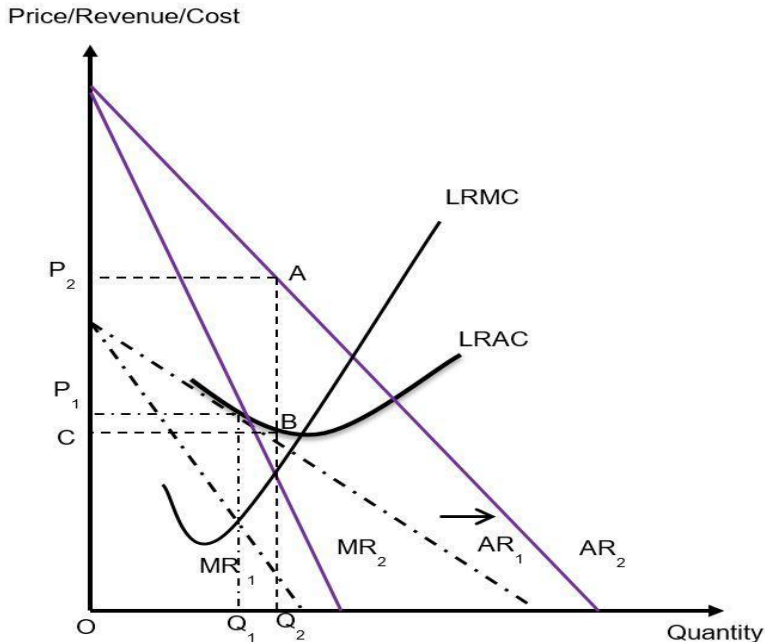
Introduction:

- **State the objective of a firm** – maximise revenue and minimize costs [maximize profits]
- Clarify that the anti-competitive/anti-trust laws likely to affect firms operating in oligopolistic market structure with high strategic and natural BTE. Context: With the anti-trust laws in place, firm's profits will fall as analysed in (a).

Revenue increasing strategies:

1. Advertising and establishing brand loyalty

- Advertisement helps the customers in their decision making process as it reduces their search cost for the product they are looking for. In these circumstances, the advertisements are extremely useful and are a low cost way of bringing all the information necessary to facilitate a buy to the customer. However, advertising can also be persuasive in nature, intended to change consumers' behavior.
- For example, when Apple promoted its smartwatch, Tim Cook said that it will "tap you out of your laziness" and remind "you to get up and move since sitting is the new cancer". They focused on the product functionality that would appeal to the consumers such as how it delivers breakthrough wellness and fitness capabilities.
- As seen in diagram below, this will lead to an increase in demand (AR_1 to AR_2) and reduce PED where AR_2 is steeper as demand has become less price elastic, increasing the profit maximising output from Q_1 to Q_2 . The higher scale of production also leads to potential cost-savings through iEOS which is represented by a movement along the LRAC curve point B and increase price from P_1 to P_2 . This will increase profits from normal to supernormal (P_2ABC).



- **Limitations**

- Advertising is costly and result in wastage of resources. If the cost of advertising exceeds the revenue generated from the higher demand for the product, the firms will see an overall fall in profits.
- The result of advertising can only be materialized in the long run as it takes time to change consumers' taste and preference and choices. As such, it defers revenue generation.
- Hence, the above strategy will be ineffective in addressing the fall in profits due to the rise in the level of competition.

2. Product/Service differentiation and product innovation

- Product differentiation increases the opportunity cost of switching by lowering the degree of substitutability of products sold by Apple, Google, AT&T and Verizon. Product differentiation can deepen brand loyalty by convincing consumers that the product is different from those offered by the rivals.
- Product differentiation by offering reward programs are effective in increasing customers' perceptions of switching cost, thus fostering customer retention as switching may involve forgoing points, expending effort and time in signing up for a new program, and learning how to redeem rewards. As a result, firms can increase switching barriers by offering attractive loyalty programs that act as a "barrier to entry" to other firms as customers may be unwilling to switch to these alternative providers despite the large number of available alternatives.
- Furthermore, service differentiation concentrated on strengthening the relationships with customers who make regular purchases can also help "keep customers from defecting" as customers may be affected by "the sunk-cost fallacy" and continue to patronize the firm because they focus on the amount of points they have already accumulated. Hence, they become locked into the program and this helps to keep the demand high despite the rise in the level of competition.
- For example, Apple has a unique operating system (iOS) which also allows iPhone users to utilize the Apple ecosystem and sync their iPhones with their other Apple devices seamlessly. Furthermore, the most dramatic change for the new iPhones, aside from better cameras that Apple always tends to offer, has the capability to make calls and send texts in emergencies when there is no cell coverage.

- As such, the above strategies will allow the incumbents to enjoy higher demand, revenue and hence profits despite the rise in the level of competition as consumers are unlikely to switch to the entrants. The effects of the rise in demand have been illustrated in the earlier diagram.

• Limitations

- It can be challenging for firms to differentiate their products to ensure that the demand for their product is price inelastic. Once a firm reveals a new idea or include a unique feature in its device, other manufacturers can also include similar features into their devices. Thus, even if a firm can differentiate its products or services, the uniqueness of its product may only be in the short run.
- This implies that firms seeking to increase its profits by increasing demand and reducing the degree of PED will have to constantly innovate and differentiate their products, either by including unique features or by differentiating their services. Otherwise, consumers may switch to the rivals in light of the rise in the level of competition brought about by the anti-trust laws.
- The above strategies can be expensive, therefore, increasing the total costs incurred by firms.

3. Diversification and partnership

- The most obvious route so far is diversifying away from the traditional product space and venturing into new markets can offset some of the loss in revenue from traditional product space by creating new revenue stream as they face a rise in the level of competition, and to a larger degree, in certain product space.
- Firms can also go into partnership to increase the overall demand for their product. For example, AT&T has agreed to use OneWeb's low Earth orbit satellites to extend high-speed broadband services to areas outside its fiber footprint in the US. Their partnership will focus on bolstering connectivity solutions for business and government customers, while also leveraging satellites to connect hard to reach cell towers across the country.
- This increases the overall demand for the firm's product, increasing in TR and profits.

• Limitations

- While this creates new revenue streams, it does not directly address the fall in revenue from the core business of the incumbent firms with the rise in the level of competition.
- Possible regulation by government for violating privacy agreements through sale of private customer information.

Cost-cutting strategies:

1. Offshoring production facilities and service centers

- Offshoring various stages of production to a country that can produce that respective stage for the lowest cost allows firms to lower their average costs, or at least mitigate the rise in costs possibly due to rising costs of research and development and advertisement.
- Additionally, firms can offshore their customer support to countries like China and India to cut cost. Offshoring certain operations to low-wage countries that also offers lower rentals have helped incumbent firms cut cost.
- Furthermore, many offshoring destinations such as the Philippines offer tax holidays, financial incentives and fewer business regulations to attract foreign direct investments and stimulate growth in industries such as offshoring and business process outsourcing. Those tax holidays and financial incentives make it more cost competitive for American firms to relocate to offshoring destinations to keep costs low.
- As such, American firms can continue to offshore their manufacturing and services to other countries to reduce the marginal and average cost of production. This will lower the total cost

of operating and hence allow these firms to maintain profits despite the rise in the level of competition.

- **Limitations**

- There will be difficulties in controlling the quality of services that end up adversely affecting the demand for a firm's product. For example, manufacturing offshoring may not guarantee that a product is strictly built according to the parent company's standards and may affect quality of the finished product. Additionally, services may be of poor quality due to communication barriers between American consumers and those providing the services at the call centers in India. As such, the reduction in costs may be insufficient and ineffective in responding to the fall in profits due to the rise in the level of competition.
- Furthermore, if the poor quality of the product and services affects the demand, profits may fall further if the TR falls more than the potential cost reductions.

2. Engage in process innovation and consolidation

- Firms can automate certain aspect of production processes as well reduce the use of office space to run their business operations to reduce the average and marginal cost of providing a service or a good, reflected by a downward shift of AC and MC.
- The pandemic has caused thousands of bank branches to temporarily close and limit hours and visits to slow the spread of the virus. However, this has changed the way the banking sector is likely to operate in the future and respond to the rise in the level of competition.
- Firms can leverage on the new normal that has introduced bank customers to new habits and digital options. As such, banks can close their branches and corporate offices to undertake branch consolidation to cut costs and leverage on their digital channels for running business since consumer behavior has changed deeply.
- In response to the rise in the level of competition, if the costs of office space could be dramatically reduced then firms would see significant rise in profits.
- Furthermore, where banking processes that are repetitive can be replaced by robotic automation and AI. It is sometimes known as "smart automation" or "intelligent automation" and thus refers to any software system that can be programmed to perform tasks that previously required the input of human intelligence to be successfully completed.
- Similarly, pharmaceutical, and medical device manufacturers have turned to automation to cut costs in response to the rise in the level of competition. This is mainly because robots can complete repetitive tasks up to 50% faster than current manual processes, enhances productivity and enabling highly skilled manufacturing, medical and laboratory staff to undertake more valuable activities.

- **Limitations**

- Automation can be costly and hurt demand for goods and services that thrives on buyer-seller relationship, particularly in the banking sector.
- Production and maintenance of artificial intelligence and robotic automation can lead to huge costs since they are very complex machines. AI also consists of advanced software programs that require regular updates to meet the needs of the changing environment.
- In the case of critical failures, the procedure to reinstate the system and recover lost codes may require enormous time and cost.
- Personal data of customers may be compromised and affect the demand for the banking services
- As such, the above strategies will be ineffective in addressing the fall in profits brought about by the rise in the level of competition.

Conclusion and overall evaluation

- Compare the strategies to decide the most appropriate strategy.

- Some possible ideas:
 - o The type of strategies depends on the amount of past supernormal profits the firms have. The lack of such profits will mean that sustained price wars are not possible
 - o Predatory pricing can be used as an immediate solution (albeit by large firms) while the firms concurrently pursue more long-term strategies such as r and d
 - o R&D may be the best strategy as there are real differences made – in terms of range and quality, and even price. That is more likely to have a more persistent effect on demand, compared to advertising

Mark Scheme

Knowledge, Application, Understanding, Analysis		
L1	- Smattering of points - Descriptive answer lacking in economic analysis on the strategies used by oligopolistic firms - Significant lapses in explanation / analysis	1-4
L2	- One- sided answer that shows adequate use of relevant economic framework - A two-sided but undeveloped answer in terms of scope of points or depth of answer analysis - Some / slight application of examples to different oligopolistic industries	5-7
L3	- A well-developed, balanced answer that has good use of economic framework that provide analytical rigour. - A well-discussed answer analysis with at least 3 good points across essay. - Answer is applied well to context of oligopolies on the pricing and output decision to demonstrate strong grasp of the content required. - Good use of examples applied to context of oligopolistic behaviour throughout essay.	8-10
Evaluation		
E1	- An unsubstantiated conclusion	1
E2	- Judgement is based on the analysis in the body.	2-3
E3	- Judgement that is well supported and evaluated from all the points highlighted. - Further insights and new (relevant) ideas are provided and substantiated.	4-5

Question 3

All countries are providing Covid-19 vaccines for free, and some advanced countries have supported patent waivers. Critics warned that lack of research and healthcare infrastructures will remain a challenge, along with growing public hesitancy towards vaccination in developing countries.

- (a) Explain the economic case for free provision of vaccines in developing countries. [10]
- (b) Discuss whether free provision is the best way to achieve an efficient allocation of vaccines in developing countries. [15]

Suggested Answer

Part (a)

Introduction:

Define market failure

- Market failure refers to the failure of the free market to allocate resources in a socially optimal and efficient manner as well as to achieve equitable outcomes.
- o Clarify free provision of vaccines – individuals can access the vaccines free of charge
- o State the reasons for the free provision of vaccines in developing countries
 - Presence of large external benefits that people ignore in pursuit of self-interest
 - Excessive income inequality in developing countries with large number of people not being able to afford the vaccines in the free market
 - High degree of imperfect information about the perceived benefits of vaccinations

1. Imperfect Information

- If left to the free market, it is likely that vaccines will be under-consumed because individuals undervalue their private benefits of being vaccinated.
- Individuals may not act in their best interests due to the lack of information about the full benefits of consuming the good.
- Imperfect Information: Individuals not fully aware of full benefits to themselves
 - Underestimate/undervalue their private benefits from being vaccinated as such, the perceived demand with imperfect information < actual demand with perfect information.
 - Consumers may be unaware of the benefits of being vaccinated, i.e, not contracting the virus and falling ill; get better protection by being fully vaccinated compared with having had COVID-19 and ability to resume activities that derive an individual higher utility, including income.
 - Since, a large number of individuals work on a daily/hourly wage basis or self-employed in developing countries like farmers, being vaccinated to prevent contracting the virus will allow them to continue to earn a stable income. According to studies, unvaccinated people who already had COVID-19 are more than 2 times as likely than fully vaccinated people to get COVID-19 again.
- Due to imperfect information, private individual in developing countries in North African, Bangladesh and Pakistan underestimate the full benefits of being vaccinated. The degree of imperfect information is likely to be larger in countries where public education levels are low and anti-vaccine sentiments are deeper given the prevalence of strong religious influence over the consumption of vaccines. As such, the perceived benefits tend to be lower than the actual benefits, resulting in a lower demand with imperfect information than that of actual demand for vaccines with perfect information.
- In the free market, private individual consumes up to only Q_e where the DD_0 intersects the SS . However, the socially optimal level of consumption is at Q_s where DD_1 cuts the SS , resulting in an underconsumption of **$Q_e Q_s$ units of vaccines**.
- **Explain DWL area:** Represented by area ABC which arises from imperfect information as the benefit forgone in not consuming $Q_e Q_s$ units exceeds the resources saved from not producing $Q_e Q_s$ units of vaccinations. Hence, **the net benefit forgone from the underconsumption of vaccines leads to allocative efficiency.**

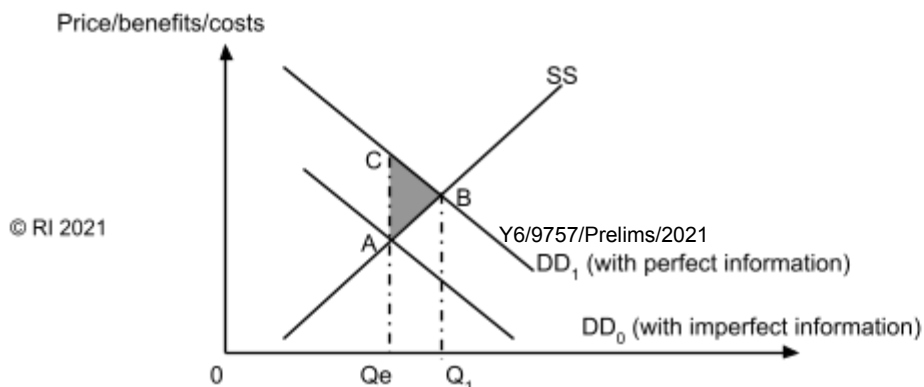
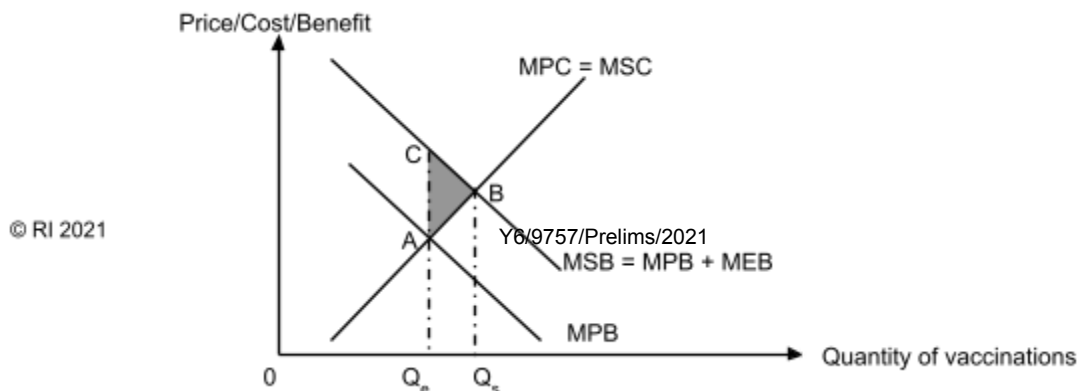


Figure1: Market for vaccines in developing countries (imperfect information)

[Turn Over

2. Presence of large positive externalities

- **Externalities:** An externality occurs when some of the costs or benefits associated with the production or consumption of a good 'spills over' onto third parties, that is, to non-participating parties (not the immediate seller or buyer).
- In the free market, due to pursuit of self-interests, consumers will only take into account of private benefits and costs such as the ability to engage in activities post-vaccination that derives them higher utility, lower risk of falling ill or hospitalization, enjoying a stable source of income due to lower rate of absenteeism. Furthermore, being vaccinated reduces the risk of contracting the new strains of virus such as the Delta variant that causes more infections and spreads faster than earlier forms of the virus that causes COVID-19. Meanwhile, the private cost is the cost of procuring the vaccines. As such, consumers will ignore the existence of external benefits of getting vaccinated such as reducing the spread of the virus that causes Covid-19 and protect those around oneself, particularly, those that are at a higher risk for severe illness from the virus like those immunocompromised individuals who cannot get vaccinated. Given the nature of the virus, the external benefits of reducing the rate at which the virus spreads through vaccination is significantly large in the context of developing countries. It is important to achieve herd immunity, particularly in informal settlements or slums in Mumbai's Dharavi, Sao Paulo's favelas, and Bangladesh's Daulatdia, where the population density is extremely high and where the virus affects large numbers.
- As a result, the external benefits of achieving herd immunity through vaccination can ensure that the informal sectors in these countries can continue to sustain themselves and contribute to economic growth.
- This means that the marginal social benefit arising from the individuals' consumption of the good (MSB) is higher than the marginal private benefit (MPB) as the benefits accrued to the third parties increases the overall benefit to the society by the amount of the MEB. Free market equilibrium point is at **point A** with **output at Q_e units**, where **$MPB = MPC$** . However, if private individuals internalize the external benefits of consuming vaccines, the socially desired output level will be **Q_s units at point B ($MSB = MSC$)**.
- **Explain DWL area ABC** – benefit from consumption of Q_e units of vaccines is greater than the cost savings from not producing Q_e units of vaccines, resulting in a **net benefit forgone by the society and hence in allocative inefficiency**.



[Turn Over

Figure 2: Market for vaccination in developing countries (positive externality)

3. Excessive income inequality

- With excessive income inequality, some people will have insufficient dollar votes. Their preference for the good is not considered by the free market.
- **Due to the excessive income inequality, there will be a lack of effective demand** since lower-income groups are unable to afford vaccines in the free market **in absence of dollar votes**. This is particularly dominant in developing countries in North Africa and South Asia where a sizeable number of people who live below the poverty line lack the ability to purchase vaccines in the free market.
- The free market will allocate resources based on the dollar votes, where $DD_0 = SS$ and produce and consumer at Q_e units of output. However, consumers who lack the willingness and ability to afford vaccines will be priced out of the market due to lack of effective demand at the prevailing market equilibrium price due to the lack of dollar votes. Meanwhile, society's welfare will improve if income is less unequal as that will cause the effective demand to increase from DD_0 to DD_1 and vaccination rate will increase to Q_s .
- **Explain DWL area ABC** – benefits from consumption of Q_e units of vaccines is greater than the cost savings from not producing Q_e unit of vaccines, resulting in a **net benefit forgone and allocative inefficiency**

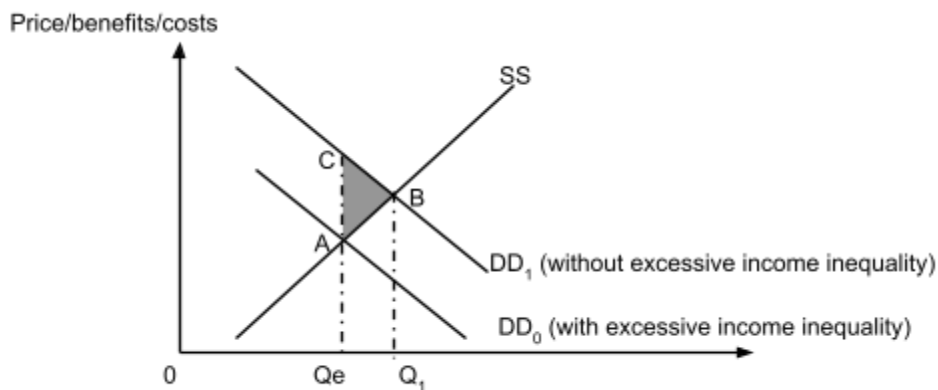


Figure 3: Market for vaccination in developing countries (with excessive income inequality)

Conclusion:

- Determine the most significant source of market failure in the context of the developing countries.
- State that the government needs to provide vaccines for free in developing countries because of the high degree of income inequality, imperfect information and large external benefits compared to the developed countries.

Mark Scheme:

Knowledge, Application, Understanding and Analysis		
L1	- Little / non-existent use of economic theory / answer that is largely descriptive knowledge with little or no economic framework applied. - Glaring conceptual errors - No examples given	1-4
L2	- Some lapses in economic analysis - Lapse in scope	5-7
L3	- Clear and excellent use of theoretical framework - Good attempt to explain the 3 reasons for free provision - Consistent use of examples to aid explanation from the developing countries	8-10

Part (b)

Approach:

- **Thesis** – Explain how the government can address the underconsumption in the developing countries by providing vaccines for free.
- **Anti-thesis** – Explain the limitations of providing vaccines for free to developing countries and explain two other policies.
- **Evaluation** – Evaluate whether free provision is the best policy in addressing the underconsumption of vaccines in developing countries and explain why. Provide suggestions to address the limitations of the policies mentioned in the analysis. May consider commenting if government intervention is likely to bring about a more desirable or undesirable outcome in the market for vaccines in the developing countries in the long-run.

Introduction:

- **Clarify free provision**
 - Providing vaccines free of charge at a price=0 to its citizens
 - State that the intervention is expected to address the underconsumption of vaccines in the developing countries caused by excessive income inequality, imperfect information and presence of positive externalities.

Thesis – Explain how the government can address the underconsumption in the developing countries by providing vaccines for free (need to provide diagrammatic explanation of how the policy works)

Explain how free provision can address the underconsumption of vaccines in the developing countries.

- Free provision means that the government supplies the goods/svcs directly to consumers usually free of charge at the point of use. In order to reduce underconsumption arising, largely from excessive income inequality, the government makes vaccines free of charge to all its citizens.
- Governments procure the vaccines from pharmaceutical companies such as Pfizer-BioNTech, Moderna and Johnson&Johnson by paying for the entire cost of procurement. This has the effect of increasing the market supply curve and shifting it to the right from SS_0 to SS_1 . As a result, the price of the vaccines to the consumers will fall to zero and this will lead to a movement along DD_0 , increasing the quantity demanded/consumption of vaccines to Q_s at $P=0$ and hence eliminating the DWL ABC created in the free market by the excessive income inequality. As such, the consumption of vaccines will not be subjected to income level.

- On average, the price of a Covid-19 vaccine is \$33, which can worth up to a month salary for many in the developing countries. With free provision, there is perfect equity, especially important for a necessity like Covid-19 vaccine in times of rampant spread of the virus in developing countries. There is a large number of people who are poor, unaware of the personal benefits and ignore the large external benefits of being vaccinated whilst living in overly crowded spaces where the virus can spread very easily.
- According to Figure 4, when price is set at zero, quantity demanded increases to Q_{FOC} , which happens to coincide with the socially optimal level at Q_S assuming that the government can accurately estimate the Q_S .
- *Alternatively, students can simply show socially efficient level of consumption takes place at Q_S when $P=0$ as DD_0 cuts the x-axis ($P=0$) without shifting the market supply curve.*

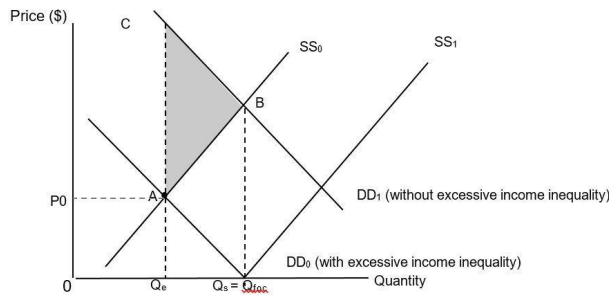


Figure 4: Excessive Income Inequality in the Market for Vaccines

Anti-thesis – Explain the limitations of free provision to developing and explain two other policies with limitations.

Explain the limitations of free provision.

- The issue is not simply about free provision of the vaccines in developing countries; it is far broader and deeper than that. The governments face a series of challenges to ensure herd immunity where the consumption of vaccines occurs at Q_S . As such, the policy of providing vaccines free will be ineffective in addressing the underconsumption to achieve the desired outcome.
- Across the globe, digital technologies, including mobile technologies, are facilitating every stage of the Covid-19 vaccine program from planning, registering for vaccination to post vaccination monitoring. However, the availability and access to internet service, electronic devices such as computers and smartphones, as well as one's ability to comprehend the procedures is lacking in developing countries. This reduces the effectiveness of the policy.
- Given the limited resources that governments have, and the alternative uses of these funds forces them to make a choice that results in an opportunity cost. As the government allocate resources to the procurement of Covid-19 vaccines, they will have less resources to employ macro stabilization policies to tackle the adverse effects of the pandemic on the economy. This, together with falling tax revenue, will result in budget deficit, particularly in developing countries. As such, providing vaccines for free for the current variant as well as future variants may become unsustainable for governments in the developing countries. The government can avoid the problem by providing it for free to those who cannot afford it, rather than to all the citizens.
- Corruption is also a major concern in developing countries. Vaccine corruption in the global south has made it more difficult to ensure an efficient consumption of vaccines despite the price of zero. In numerous low- and middle-income countries, politicians and high-ranking officials have abused their power to cut the vaccine queue. As such, some of the primary beneficiaries of the current government vaccination programs have been the professionals working in those industries and the richer individuals. However, the poor citizens may continue to remain unvaccinated.

Explain how waiving patents can address the underconsumption of vaccines in the developing countries.

- Patents serve as an artificial entry barrier - giving firms exclusive right to produce the patented medicine. The market is oligopolistic as the 10 largest drugs companies control over 1/3 of the market. They are also likely to possess monopoly power over the drugs developed - being duly protected by patents. At the firm's profit-maximising equilibrium, it can restrict output to set a high price. Its output is lower and price is higher than the allocatively efficient level where $P=MC$. Thus the under-production leads to a deadweight loss where for every unit under-produced, $MSB > MSC$. Thus, the private companies' profit-seeking motives is in conflict with society's goal for welfare maximisation.
- Waiving patents could allow vaccine technology to be shared more easily, lowering the barriers to entry as generic or otherwise non-licensed manufacturers could begin production in the countries considered to have the production capacity to do so such as India and Brazil.
- Pharmaceutical companies such as Premas Biotech, Bharat Biotech, Cadila and Eurofarma, can copy the "recipe" for the vaccination without incurring high R&D costs themselves.
 - This lower cost of production of Covid-19 vaccines, increasing the market supply of vaccines from SS_0 to SS_1 .
 - This lowers the price of these vaccines to P_1 , that is below the prevailing average price of \$33, hence increasing the quantity demanded of vaccines to Q_s . The lower price makes it more affordable for the needy, addressing the underconsumption caused by excessive income inequality as well as large external benefits since the lower price will incentivise private individuals to internalise the large external benefits and consume at the socially efficient level.
- As such, a temporary suspension of intellectual property rights related to Covid-19 vaccine will allow developing countries, who have been previously locked out, gain access to the vaccines.

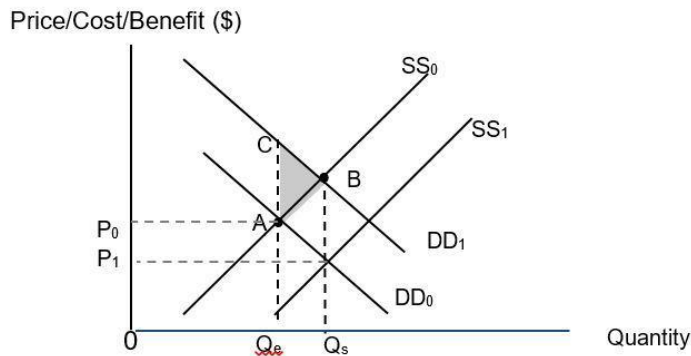


Figure 5: Market for Vaccines

Explain the limitations of waiving patents.

- Removing patents may lead to an even higher price in the interim as firms seek to recoup their cost incurred before the entry of the generics. *This will make it more expensive for governments to procure these vaccines, worsening the negative effects of providing the vaccines for free to all its citizens.*
- Waiving patents is likely to disincentivise innovation at a crucial time and ignores the true cause of supply constraints – difficulties in expanding manufacturing capability. Pharmaceutical companies have generally opposed these suggestions, arguing that any suspension of patent protections would have damaging effects on innovation.
- Hence, with removal of patents, pharmaceutical companies will cut back on future innovations to deal with possible future variants that are likely to be more deadly.

- Manufacturing capability and regulatory clearances also present significant challenges for any manufacturer to overcome. For example, AstraZeneca has had well-publicised yield issues, and appropriate sites for production have proven hard to come by: only three production sites identified by AstraZeneca had received regulatory approval from the European Medicines Agency to produce the vaccine as of last month (one each in Belgium, the US and UK). Some countries have paused rollouts of certain vaccines based on health concerns by regulatory authorities. Similar issues would apply in developing countries, and for any manufacturer seeking to supply to those countries.
- As such, it is unlikely that many would see expanding production at the expense of safety as a sensible trade-off, given it could risk undermining public confidence in the safety and efficacy of the vaccines being produced, and negatively impact vaccine uptake rates as a result. Overall, the underconsumption of vaccines may remain.

Explain how regulation can address the underconsumption of vaccines in the developing countries.

- Legislation such as mandatory vaccination should be imposed to ensure that consumption takes place at Q_s , eliminating the DWL arising from imperfect information and private individuals ignoring the external benefits.
- To avoid paying hefty penalties for flouting the regulation, individuals will be forced to internalize the external benefits. As such, this will increase the MPB closer to MSB, resulting in a socially efficient outcome.

Explain the limitations of regulations.

- Some individuals may view the policy as highly intrusive and evasion of their rights to choose to be vaccinated.
- The policy is costly, as it requires additional funding on enforcement and monitoring. This is particularly more significant in developing countries where certain areas are not accessible and deploying such a taskforce will be difficult.
- If penalties of breaking the law is not sufficiently high, then it will be very difficult to ensure that consumption takes place close to the socially efficient level. If individuals can get away with paying the penalties then there is no incentive to internalize the external benefits, more prevalent in developing countries where individuals can get away with paying government officials bribe if they are caught flouting rules.

Explain how education and campaigns can address the underconsumption of vaccines in developing countries.

- Public service announcements can overcome the information failure associated with vaccines so that people will appreciate the full private benefits of being vaccinated and importance of achieving her immunity. Positive advertising using TV personalities to create awareness on the full benefits of attaining vaccines and influencing the mind-set of individuals is a very effective tool to increase demand for vaccine.
- As such, public education in terms of campaigns and advertisements through the many different media may equip consumers with more accurate information and lead them to increase the demand with imperfect information closer to demand with perfect information for vaccinations and thus lead to the socially optimal level of consumption at Q_s .
- In Bangladesh, the government creates videos to encourage frontline workers – such as healthcare workers, teachers and police officers to instruct others in their community to self-isolate, get vaccinated and call a doctor on the helpline using a method known as “edutainment”. It is a video-based content seeking to educate the Bangladeshis whilst entertaining them. This has been largely effective in communicating the actual benefits of being vaccinated through social media using YouTube and Facebook.

- Campaigns/education to overcome information failure may **be more effective** in the long-run as it addresses the root cause of the problem and it may also be **more desirable** as it is less costly than free provision of vaccines.

Explain the limitations of education and campaigns.

- It takes a long time to change the mindset of people against their previously held beliefs about vaccines. This is particularly more difficult in developing countries where there is a strong correlation between educational attainment and receptiveness to information. As such, these campaigns may be largely ineffective in increasing demand for vaccines and eliminating the underconsumption arising from imperfect information.

Conclusion and Overall Evaluation

- **Stand** – State whether free provision of vaccines is the best way to address the underconsumption of vaccines in developing countries.
- **Explain stand**
- Compare the policies based on the yardsticks – effectiveness, desirability, sustainability, feasibility by assessing short-run versus long-run responses.
- **Insightful comments** – Application to the COVAX facility, ways of financing the various policies in the context of developing countries [foreign aid, raising wealth taxes].



- Given that the elderly and children most susceptible to contagious diseases because of low immunity and the excessive income inequality in developing countries, free provision of vaccines is the best way to address the underconsumption.
- However, for others that can afford the vaccine, a smaller degree of indirect subsidy could also be granted because of the large positive externality. Ensuring a healthy community is very desirable because if labour force is healthy, it can lead to increases in output and services in the economy, contributing to actual and potential growth; it will also lower medical cost and opportunity cost if the people in the country are healthy.
- There are other factors that contribute to the high inequality in vaccination rates and hence an inefficient allocation of vaccination in developing countries. Healthcare and educational systems are still highly unequal in developing countries, in other words, educational inequality is very much deeply entangled with other problems- disparities in income, gender and of course health. Such inequalities are often overlooked and are not addressed effectively and appropriately by policymakers. As such, this contributes to the high degree of underconsumption of healthcare services such as vaccines. Therefore, provision of vaccines do not necessarily translate into equal access to vaccinations.
- Nonetheless, free provision is the best way to improve the allocation of resources but public education also needs to be used to correct for other sources of market failure like the lack of information in the industry.

Mark Scheme:

Knowledge, Application, Understanding, Analysis		
L1	- Answer lacks relevance - Lots of glaring conceptual errors throughout	1-4

	- <i>Journalistic approach, with no theoretical framework</i>	
L2	- <i>Lapses in scope provided</i> - <i>Lapses in depth</i> - <i>Some lapses in details in explanation</i>	5-7
L3	- <i>Good scope provided</i> - <i>Good use of theoretical framework</i> - <i>Displays fluency and clarity throughout the answer</i> - <i>Good attempt in explaining and analysing the workings of every policy with clearly illustrated diagrams.</i> - <i>Excellent and consistent use of examples from the developing countries in the context of Covid-19 vaccines.</i>	8-10
Evaluation		
E1	- <i>Judgement is not explained.</i>	1
E2	- <i>Well-reasoned judgement on whether free provision is the best way</i> - <i>Judgement is based on economic analysis and/or empirical evidence</i> - <i>Judgement is explained</i>	2-3
E3	- <i>Judgement is based on strong economic analysis, empirical evidence and contextual application, as well as a well supported conclusion</i>	4-5

Question 4

Entering the second quarter of 2020, Americans' fears of the pandemic were ramping up. In response, the country went on an unprecedented saving spree.

Source: Federal Reserve Bank of Kansas City, 29 April 2021

(a) Explain why household savings could increase in a country.
[10]

(b) Discuss whether a higher rate of household savings is beneficial to a country.
[15]

Suggested Answer

Part (a)

Introduction:

- Define household saving
 - o Fraction of disposable income that is not consumed by households.
 - o $S = Y - T - C$

1. Weak state of the economy

- Expectations of future income
Due to an expected fall in future income (bad news on prospects of recovery from pandemic or vaccine inefficacy etc.), households expect unemployment to increase or pay cuts down the road. This leads to a fall in autonomous consumption as there is a corresponding increase in precautionary household savings in order to smooth out consumption over a longer period of time to maintain SOL.

- Expectations of price changes/Deflation
Due to a fall in AD, which could be due to a fall in net exports (or any other relevant AD component), expectations of deflation may set in. This could cause a fall in consumption as consumers expect lower prices in the future which will allow them to buy more goods and services. This will lead to a rise in household savings. Furthermore, if the nominal interest rates remain constant, the real value of savings will increase with deflation. Real value of saving increases due to fall in GPL.

2. Government policies

- Contractionary Monetary Policy: Increase in interest rates

Due to a high inflation, governments will reduce an overheating of the economy by implementing contractionary monetary policy. Central banks will increase nominal interest rates. Assuming *ceteris paribus*, this will increase the real interest rates in the economy, which will increase the returns to savings to savers. Savings will rise. As the returns to savings increase, this will increase the opportunity cost of consumption, hence consumption will fall, and households will be incentivised to save more.

Furthermore, as the cost of borrowing is higher due to higher interest rates, big ticket items become more costly to finance and households are disincentivised from consumption. Thus, household savings will rise.

- Expansionary Fiscal Policy

Due to a recession, governments implement expansionary fiscal policy to stimulate economic growth. This could be due to an increase in government expenditure or a fall in personal income taxes, which would cause an increase in disposable income. This rise in disposable income would allow for more to be saved, which could increase in the level/rate of savings.

The rise in G & T would also lead to multiple rise in national income, in which withdrawals in the form of taxes, savings, and imports would also rise. This would cause an increase in the overall level of savings in the economy as national income increases for given MPS.

OR Due to the Ricardian Equivalence theory, rational and forward-looking households will increase their current level of household savings due to a current government budget deficit. These households predict that there would be a rise in income taxes in the future to offset the budget deficit in the current period. Households do this to smoothen consumption in the current and future periods. Hence, household savings will rise.

- Introduction of compulsory savings schemes/increase in compulsory savings rate

If government increases the rate of compulsory savings due to expectations of rising cost of living (rising healthcare costs, longer life expectancies), and more challenging economic conditions in the future, this will increase the rate of savings in a country.

3. Level of savings due to a rise in income

- Similar to expansionary FP argument above

- The rise in any component of AD will lead to multiple rise in national income through the multiplier process, which will lead to rise in leakages which includes savings.
- Hence, as the country experiences economic growth, there will be a higher level of savings.
- Alternatively, as household income rises, savings rate (MPS) will rise as there is more discretionary income to save (after necessities have been purchased).

4. Other acceptable factors

- Due to changes in demographics such as an ageing population, there would be a change in the economy's rate of savings in the long run. Retirees tend to save more and spend less due to lowered incomes, thus affecting the overall MPS and MPC and the overall change in savings.
- Improve access to banking facilities in developing countries

Mark Scheme:

Knowledge, Application, Understanding, Analysis		
L1	Conceptual errors and inaccuracies. An answer that is irrelevant. A descriptive response. Only 1 determinant explained	1 – 4
L2	Some gaps and lapses in explanation. Explanation lacks rigour for some determinants. Lacking scope of analysis Largely theoretical with limited application of examples	5 - 7
L3	Consistent use of examples. Sufficient depth with few conceptual errors 3 well-explained determinants grounded in economic theory.	8 - 10

Part (b)

Introduction:

- Increase in savings implies fall in consumption, ceteris paribus
- Whether it is "beneficial" to a country all depends on impact on its the macroeconomic indicators and SOL.

Body

Thesis: A higher rate of household savings is not beneficial to a country

1. The country may not be able to achieve actual economic growth and low unemployment due to a fall in consumption

Holding income constant, an increase in savings rate causes consumption to fall. This will cause the AD to fall and unplanned investment occurs, causing firms to step down production in the next cycle. The fall in autonomous C sets off successive rounds of falls in induced C, leading to a more than proportionate fall in RNY by the reverse multiplier process, which is based on the fact that one man's spending creates another man's income. The process will stop when the sum of

withdrawals is equal to the initial fall in autonomous C. This will lead to a fall in national income and economic growth. Since firms produce less, derived demand for labour falls. This causes an increase in demand-deficient unemployment

2. The country may face a fall in standard of living

Assuming population remains constant coupled with a fall in RNY, there will be a fall in purchasing power of households, reducing their consumption of goods and services and hence satisfaction of needs and wants, leading to a fall in material SOL.

OR Due to a rise in savings, there is a fall in the level of consumption in the current period. This will cause a fall in material SOL in the current period.

3. The country may not be able to achieve price stability due to a fall in consumption

For countries facing deflation, the fall in AD might also cause GPL to fall further, setting off the deflationary spiral where consumers further reduce C in expectations of further falls in prices, leading to further falls in GPL and RNY.

4. The country may worsen a worsening BOP because of a worsening of capital and financial account

With an increase in savings supply of loanable funds increases, which will lead to a fall in interest rates. This will lead to an outflow of hot money as domestic and foreign investors now seek higher rate of returns elsewhere. This worsens the capital and financial account position in the BOP.

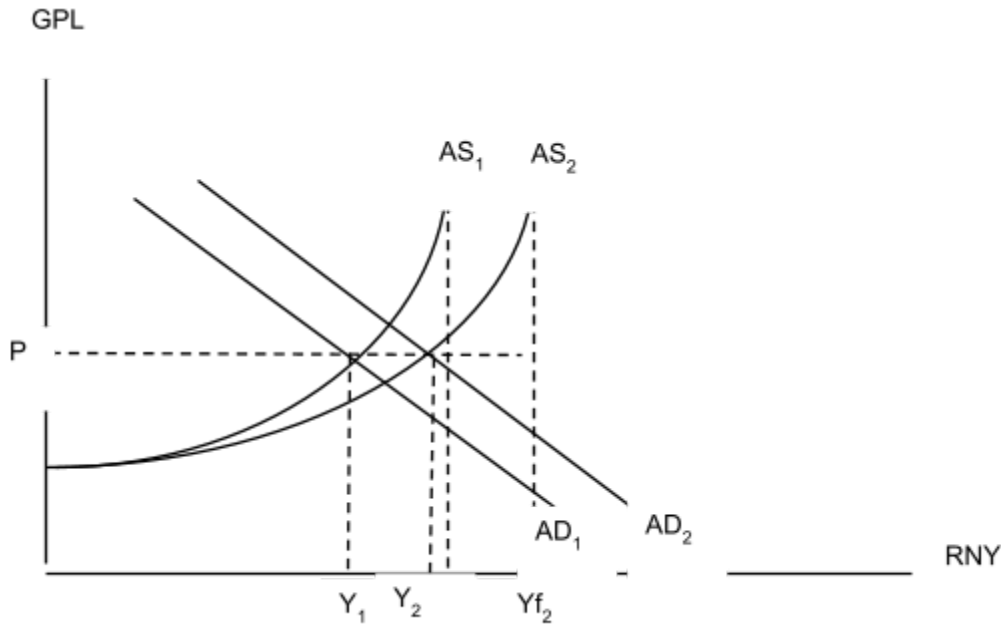
If the outflow of hot money is sustained and leads to depreciation and further capital flight, this might lead to exchange rate instability. (This must be linked to the macroeconomy. It is not necessary to explain the effects of a depreciation).

Anti-thesis: A higher rate of household savings is beneficial to a country

1. There will be an increase in sustained economic growth in the longer run

With the increase in savings rate, supply of loanable funds increases and hence interest rates fall. This will lead to an increase in investments based on the MEI theory. The MEI refers to the expected rate of return of an additional unit of investment while the rate of interest refers to the cost of borrowing (assume that firms borrow from banks to invest). As interest rates fall, cost of borrowing falls, and the number of profitable investments projects with MEI greater than interest rates will increase. Thus, level of investment will increase.

Investments increases AD and through the multiplier process, there is an increase in real national income and actual growth. In the long run, with the accumulation of capital due to investment, it results in an increase in factor quality and quantity, increasing AS, shifting the AS curve outwards. This promotes potential economic growth. Coupled with actual growth, the economy will achieve sustained economic growth.



2. There will be an increase in the standard of living in the future/long-run

As the productive capacity of the economy increases, it also allows for an increase in the production of consumption goods in the future, increasing material SOL. From the household's perspective, there will be a greater level of consumption in the future due to high level of savings now based on the assumption that the real interest rate in the economy is positive.

3. There will be an improvement in price stability

The expansion of productive capacity allows increases in AD to be accommodated without inflationary pressure, hence allowing for not only sustained growth but also reduces the extent of demand-pull inflation.

OR Due to a fall in consumption, there will be a fall in inflationary pressures in an economy that is producing on the vertical range of the AS curve. This will help promote price stability in the economy.

4. There will be an improvement in the Balance of Trade

With a reduction in consumption of goods and services including imports, trade balance and the current account position improves. OR With a fall in RNY due to a fall in C, there is a further fall in willingness and ability to consume imports, assuming normal goods, further improving CA.

Conclusion and Overall Evaluation:

Whether household savings will be beneficial depends on:

- Characteristics of economy: Cd as a % of AD compared to M OR C vs I
- State of economy: Increase in I might not be materialise because of pessimism and expected rates of return to investments remain low, hence there is no capital accumulation and LR EG. In

addition, *ceteris paribus* might not hold, other components of AD may be falling at the same time, worsening the recession.

- Root cause of the rise in savings: A rise in household savings could be due to expectations of deflation. This could mean that there would be a significant fall in C and a minimal rise in I. This is less beneficial overall to the economy. However, if the rise in household savings is due to economic growth and consumer and firms' expectations are healthy, the rise in I could be greater than a fall in C. This would mean that the benefits would outweigh the costs overall.
- Time period: In the short run, it is likely that the effects of an increase in household savings will be less beneficial. However, this may not be the case in the long run with potential economic growth.
- Mitigating policies / Policy Suggestions: Government may also implement expansionary demand-management policies or SR cost-cutting measures to address the negative impact that arose due to the rise in household savings.

Mark Scheme:

Knowledge, Application/Understanding, Analysis		
L1	• Answer attempts to present some arguments on how savings impact the macroeconomic aims / SOL of a country. • Analysis is largely undeveloped and contains inaccuracies in content or is one-sided without scope. • No contextual knowledge to support arguments. Answers are purely theoretical. Examples have not been used. • Lack of AD-AS analysis	1-4
L2	• Answer presents well-balanced arguments on how savings impact the macroeconomic aims / SOL of a country, although the analysis may not be fully developed. • Analysis is developed but may have gaps in arguments. • Some attempt to contextualise using examples • One-sided answer	5-7
L3	• Answer demonstrates a good understanding of how savings impact the macroeconomic aims / SOL of a country • Economic analysis is well-developed throughout with and argument is coherent. • Answer makes use of real-life examples to support arguments. • Answer covers scope of short-run and long-run aims	8-10
E3	- A well-justified stand that considers unstated assumptions and provides a new perspective	4-5
E2	- Answer makes a stand that attempts to use economic justification.	2-3
E1	- Answer makes an unjustified stand.	1

Question 5

- (a) Using the circular flow of income, explain how the effects of an increase in exports on the equilibrium level of national income may differ between a small and open economy and a large and less open economy.
[10]

- (b) To what extent is a country's macroeconomic priority dependent on the openness of its economy?
[15]

Suggested Answer:

Part (a)

Introduction

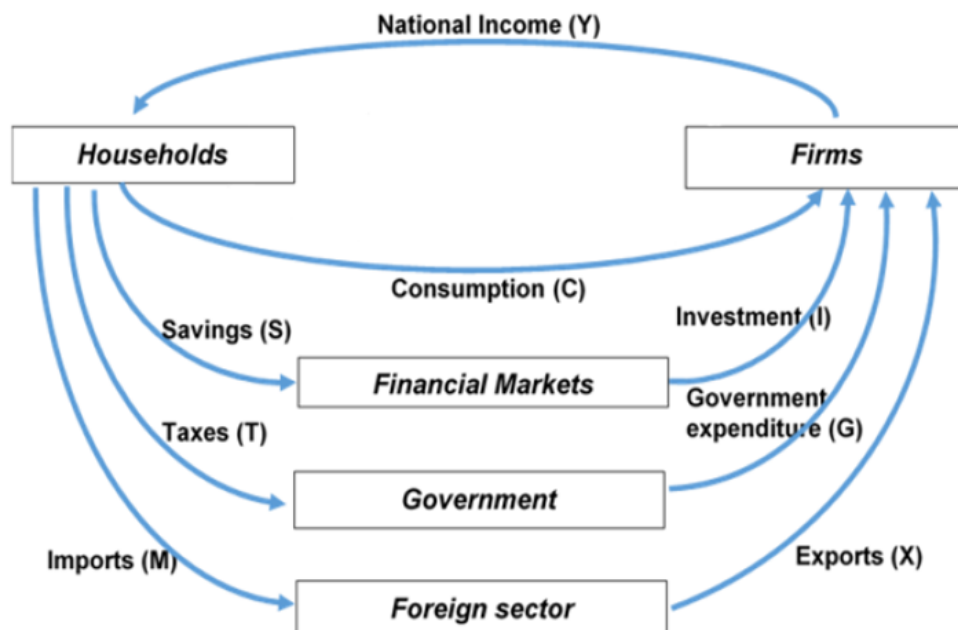
- The circular flow of income (cfi) shows the flow of output, income and expenditure in a 4-sector economy consisting of households, firms, the government, and foreign sectors.

Development 1: Explanation of how circular flow of income works

- The uppermost flow represents national Y (wages, rents, interests, and profits) which are accrued to households in return for factor services.
- The inner loop linking households and firms is the central income and output generating mechanism.
- Income leaves the circular flow as withdrawals (W) consisting of S, T and M.
- Income enters the cf as injections (J) consisting of I, G and X.
- National Y is in equilibrium when Total Injections = Total Withdrawals

(Note: Students are expected to draw and explain the circular flow of income diagram).

Figure 1: The Circular Flow of Income Model



Development 2: Explain how the effects of an increase in exports on the equilibrium level of national income may differ between a small and open economy and a large and less open economy.

- Small and open economies (SOE) have a high total trade to GDP ratio whereas large and less open economies (LLOE) have a smaller total trade to GDP ratio.

Explain the k-effect from an increase in X using injection-withdrawal approach.

- Suppose both types of economies experience an increase in X of 1m.
- Total injections exceed total withdrawals causing RNY to increase. Since expenditure=income, household income in both types of economies increase by the same amount of 1m initially since domestic firms earn revenue from the foreign sector.
- Difference mpw is higher in SOE due to higher mpm (lack of resources leading to higher import dependence) unlike LLOE. Assume mpw=0.5 for SOEs and MPW is 0.25 for LLOE. 500k is withdrawn from SOE as savings, taxes and imports compared to 250k for LLOEs.
- Hence 500k is recirculated back into the SOE as induced consumption in the inner loop by households (mpc= 0.5) whereas 750k is recirculated for LLOE (mpc=0.75).
- As expenditure = income, RNY increases by only 500k in SSOE whereas it increases by 750k in the LLOE.
- From the above increase of RNY, a further 50% is withdrawn for SOE whereas for LLOE only 25% is withdrawn. Once again, a higher proportion of income is recirculated back as induced consumption for LLOEs creating more additional income in the inner loop.
- The process of expenditure and income creation stops when the withdrawals have increased by 1m which is equal to the initial injection.
- RNY increases by 2m in the SOE whereas for the LLOE it increases by 4m because the size of the k ($1/\text{mpw}$) is larger for the LLOE.
- The above analysis assumes both types of economies have the same absolute increase in X ceteris paribus.

Development 3: Relative importance of X

- The relative importance of the X as injection is greater for a SOE compared to an LLOE. Although size of k is smaller in an SOE, the proportion of X/GDP is likely to be higher as well.
- Hence the impact on RNY will still be significant for a SOE even though the k size is small. On the other hand, X/GDP is relatively lower for a LLOE so the relative importance of X as an injection on RNY is less significant compared to other injections such as I or G even if the k size is larger.

Mark Scheme:

Knowledge, Application, Understanding, Analysis		
L1	<i>Conceptual errors and inaccuracies. A descriptive response.</i>	1 – 4
L2	<i>Some gaps and lapses in explanation. Circular flow model not used effectively, well explained or drawn. Superficial comparison made between small and large and less open economy. Use of AD-AS framework to explain k effect instead of injections and withdrawals approach.</i>	5 - 7

L3	<i>Good explanation of circular flow model and k effect using injections and withdrawals. Comparison of k effect between SOE and LLOE is thoroughly explained and well-justified.</i>	8 - 10
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Part (b)

Introduction:

- Define degree of openness as total trade (X+M) to GDP ratio.
- The four macroeconomic priorities are low unemployment, low inflation, economic growth and a healthy balance of payments.

Thesis: Openness determines macroeconomic priority.

Development 1: SOEs

T1: Openness to external demand shocks makes *healthy BOP* the priority.

- Openness could increase the importance of external balance for SOEs; i.e. avoidance of CA or BOP deficits.
- For e.g., When there is a fall in X demand due to global financial crisis or pandemic.
 - o Fall in profit of domestic firms retrenchments.
 - o Fall in AD reverse k effect fall in RNY slow or negative EG fall in SOL.

To avoid the above costs, the priority should be improved BOP by increasing X via ER MP for e.g., a depreciation/zero appreciation

increase X-M Improves BOT if $PED_x + PED_m > 1$ healthy BOP (avoidance of large and persistent deficits).
In addition, when (X-M) increases Increase AD Increase RNY reduce cyclical Un
improve SOL.

or

T2: Openness to external supply shocks makes *low inflation* the macroeconomic priority.

- SOEs are susceptible to imported cost-push inflation due to their dependence on imported raw materials and intermediate products.
- Increase in P_m increase in cost of production fall in AS (shifts up) increase in GPL.
- Fall in RNY, possible increase in unemployment as less workers are demanded to produce lower output levels fall in SOL.
- High inflation could have secondary effects on investment and export competitiveness, worsening the BOP.

To avoid the above problems, low inflation could be prioritised via a gradual appreciation of the currency. As this happens, P_m falls, causing the AS to shift down and GPL to fall, moderating the inflation.

Although in the short-run it conflicts with the goal of healthy BOP, assuming MLC condition holds, in the longer-term if domestic inflation < foreign inflation, (X-M) and FDI increases, improving BOP. AD and AS increases achieving actual and potential growth (sustained EG).

Hence SOEs should target low inflation as macroeconomic priority.

Development 2: LLOEs

T3: Lack of openness suggests EG is the priority

- In contrast to SOEs, LLOEs are less exposed to external shocks. Hence, they can pursue EG directly via appropriate policies e.g. expansionary FP due to their large k size.

Elaborate on the some of the benefits of EG.

- Easier for governments to redistribute income.
- Improvement in material and non-material SOL.

Or

T4: Lack of openness suggest low unemployment is the priority

- Large economies have a large population size. Unemployment might be a more difficult problem to manage. Since domestic market is large and they are not dependent on X and FDI to sustain employment, the priority could be maintaining low unemployment.

Elaborate on the benefits of low unemployment.

- Higher productive efficiency.
- Less strain on the budget.
- Improvement in material and non-material SOL.

Anti-thesis: Openness does not determine macroeconomic priority

AT1: Macroeconomic problems caused by domestic sources (i.e. not due to external shocks or openness).

- Low inflation should be the priority if there is high inflation due to domestic sources such as wage-push or demand-pull inflation caused by speculative investments.

Or

- Low unemployment should be the priority if there is high unemployment due to structural rigidities in the labour market e.g. presence of strong-trade unions, workers are overspecialised etc.

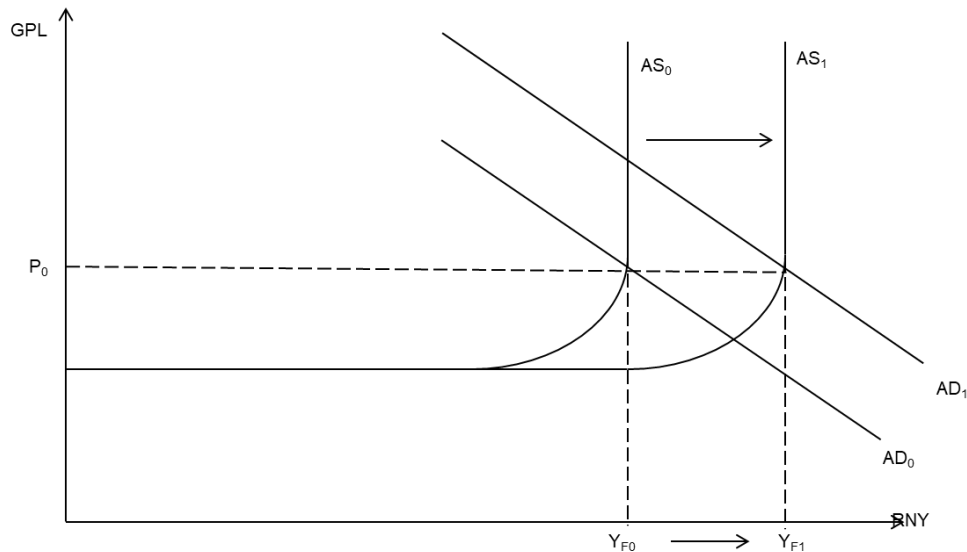
Or

- Economic growth should be the priority if there is slow or negative growth due loss of confidence or a pandemic within the country.

(Elaborate on the benefits of the any of the alternative goals above).

AT2: Sustained EG is the goal in the long run regardless of openness.

- All economies aspire to achieve sustained EG.
- Both AD and AS increases Actual and potential growth results in price stability (low and stable inflation).
- Unemployment will be low as the economy operates near Y_f .
- BOP improves due to price stability.
- Sustained EG is also a prerequisite for sustainable growth.
- The government's budget improves allowing for income redistribution and inclusive growth.

Fig 2: Sustained Economic Growth**Conclusion and Overall Evaluation:**

- Openness determines susceptibility to external shocks and raises the significance of the BOP. Lack of openness results in the opposite.
- The degree of openness does determine the macroeconomic priority because it affects the freedom governments have in choosing their priority
- However, in the long run all economies aspire to achieve sustained EG hence the degree of openness is less relevant in determining the priority.
- While all economies suffer from macroeconomic problems due to domestic sources, the severity of problems from external shocks far exceeds domestic shocks for small and open economies as compared to LLOEs implying that openness is the main determinant of the priority for SOEs.

Mark Scheme:

Knowledge, Application, Understanding, Analysis		
L1	<i>An answer which has no linkage to openness Little / non-existent use of economic theory / framework Conceptual errors</i>	1 – 4
L2	<i>Some lapses in explanation either in linking the priority to openness or in the elaboration of the benefit of the priority Balanced with some attempt to link openness to the priority. Some use of AS/AD framework in the explanation One-sided response</i>	5 - 7
L3	<i>A balanced and well-developed answer with good linkage between openness and priorities. Good scope: A total of 3 main ideas.</i>	8 - 10

Evaluation		
E1	<i>A weakly substantiated judgement.</i>	1
E2	<i>Judgement is backed by reason and well-supported.</i>	2 - 3
E3	<i>Judgement is well-reasoned and insightful. Multiple perspectives are adopted</i>	4-5

Question 6

Supply-side policies can (i) improve the quantity, quality and mobility of factors of production, and (ii) increase efficiency, and remove barriers to competition in product markets.

- (a) Explain how such supply-side policies can improve Singapore's balance of trade.
[10]
- (b) Discuss the factors that determine the severity of potential challenges that might arise when a country loses its competitiveness.
[15]

Suggested Answer:

Part (a)

Introduction

- Clarify balance of trade – $TR_x - TE_M$
- Clarify supply-side policies - policies aimed to increase the potential output of an economy and hence contribute to non-inflationary economic growth, as well as an improvement in balance of trade at the same time.

Body: Explain any 3 policies/measures

1. Policies to improve the quantity, quality and mobility of factors of production

(a) Infrastructural development

- Transport and communications – e.g. extension of MRT network, North-South Expressway in Sg
Increase accessibility and speed of getting to workplace and hence productivity.
- High-speed Next Generation Nationwide Broadband Network Improve connectivity and workers' productivity.
- Business parks for various industry clusters e.g. Biopolis, Jurong Island, Data Centre Park, Seletar Aerospace Park.

(b) Human capital development

- Expand capacity and enhanced funding support of the Continuous Training and Education (CET) in key growth areas identified by the government such as aerospace, precision engineering and process manufacturing.
- SkillsFuture Credit aimed to encourage individuals to take ownership of their skills development and lifelong learning. Introduced in 2015, all Singaporeans aged 25 and above will receive an opening credit of S\$500. To further encourage Singaporeans to reskill and upskill to seize opportunities in the future economy, a one-off SkillsFuture Credit top-up of \$500 was provided on 31 December 2020.

- Workforce Singapore (WSG) opened up more than 2,000 training places under its Adapt & Grow initiative to support five sectors hit hard by the COVID-19 outbreak.
- Under the Professional Conversion Programme (PCP), course fee subsidies are given to help professionals, managers, executives and technicians (PMETs) to undergo skills conversion and move into new occupations or sectors that have good prospects and opportunities for progression.

(c) R&D grants to spur innovation

- In 2000, Agency for Science and Technology (A*STAR) was set up to build up Singapore's capabilities in science, engineering and technology.
 - A \$25 billion plan charting the country's research landscape was launched in December 2020 under the Research, Innovation and Enterprise (RIE) 2025 plan.
- Encourage product and process innovation to improve both quality and quantity of factors of production and/or quality of product.

(d) Fiscal and governmental reforms

- In Singapore, corporate income taxes were cut from 20% to 18% in 2008 and again to 17% from 2010 onwards to encourage investment. As after-tax profitability increases, firms become more willing and able to invest as they now can retain more profits.
- Cutting income taxes encourages work effort and increases the quantity/quality of labour supplied, assuming substitution effect outweighs income effect. It also helps to attract / retain high-skilled talent.
- Reduction of red tape, removal of other impediments to investment and risk taking e.g. establish intellectual property rights (IPRs). Increase firms' incentive to undertake investment projects for capital accumulation.

2. Policies to improve efficiency and remove barriers to competition

- Free up the markets by removing distortions of current policies on prices and incentive. Encourage private enterprise and improve market efficiency.

(a) Privatisation

- Transfer of a state-owned monopoly run by the government to the private sector or introduce private services into the public sector. Introduction of competition and profit-motive to reduce x-inefficiency.
- E.g. Privatisation of SMRT, Singapore Airlines, Singtel. Note that even though these goods are not exported, it helps to reduce cost of production for export-oriented firms via cheaper transport, utilities, communication costs etc.

(b) Deregulation

- Governments remove, reduce, or simplify restrictions on firms. Fewer and simpler regulations, giving out of more licences. Increase the level of competitiveness by encouraging the efficient operation of markets and spurring innovation, encouraging productive and dynamic efficiency.
- E.g. Open Electricity Market (OEM), TPG becoming fourth telco in Singapore. Note that even though these goods are not exported, it helps to reduce cost of production for export-oriented firms via cheaper transport, utilities, communication costs etc.
- Also applies to trade liberalisation

(c) Flexible labour policies

- Encouraging female labour participation by having more maternity and childcare benefits, and raising the retirement age helps to increase competition in the labour market and possibly increase the supply of labour. Downward pressure on wages and improvement in productivity to expand productive capacity and decrease unit COP.

(d) Reduction of government spending

- Cutting back government spending e.g. firms' subsidies that distort markets to free up resources for the private sector which, subjected to the discipline of market competition, tend to be more efficient than the public sector Improves productivity.

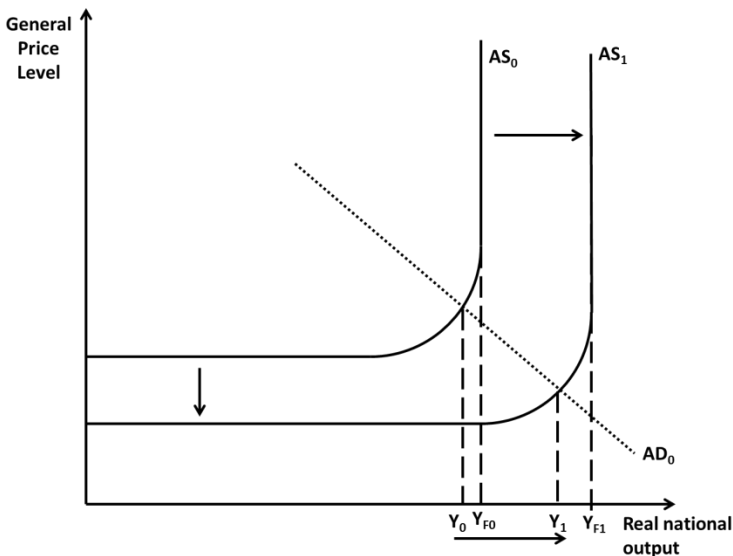
3. Explain how policies work to improve BOT

(a) Increase non-price competitiveness of domestic goods and services

- With the use of better technology and machinery for product innovation, quality of domestic goods and services improve.
- As domestic goods and services cater better to consumers' taste and preferences, demand for exports rise and becomes more price inelastic Revenue from exports increases even when prices increase.
- For instance, Singapore is home to renowned pharmaceutical companies such as GlaxoSmithKline, MSD and Roche, driving our exports of pharmaceutical products as the coronavirus pandemic prompted worldwide stockpiling of drug ingredients.
- Similarly, expenditure on imports will fall as consumers switch to domestic goods and services, and substitutability to imports fall.
- For instance, Singapore has developed home-grown brands such as Prism+, Secretlab, Razer that has reduced our reliance on imports in these goods and services.

(b) Increase price competitiveness of domestic goods and services

- The supply-side policies mentioned above help to expand productive capacity from Y_{F0} to Y_{F1} as represented by an outward shift from AS_0 to AS_1 , reducing the likelihood of supply bottlenecks and hence **reducing demand-pull inflation**.
- In addition, the improvement in factor quality and quantity help to reduce unit COP as less factor inputs are required to produce each unit of output, and factor input prices fall as they become more abundant, **reducing cost-push inflation**.



- Due to lower relative inflation rate compared to trading partners, goods that are produced domestically are now relatively cheaper than trading partners' goods. Singapore as foreigners switch from other trading partners' goods to our exports, increasing demand and hence export revenue.

- OR A fall in P_x results in an increase in quantity demanded, assuming $|PED_x| > 1$ with many substitutes from foreign counterparts, the more than proportionate increase in quantity demanded of exports leads to an increase in export revenue.
- Stable prices at home will reduce the tendency for households to switch to import substitutes because domestic goods and services are price competitive $\downarrow TE_M$, assuming that they are substitutes.

Mark Scheme:

Knowledge, Application, Understanding, Analysis		
L1	Mere listing of policies or largely irrelevant answer e.g. policies linked to other macroeconomic goals. Glaring conceptual errors	1 – 4
L2	Lacks depth (undeveloped explanation / lacks rigour in analysis despite some use of economic framework). Lacks breadth – Less than three relevant policies well-explained. Limited use of real-world examples relating to Singapore.	5 - 7
L3	Well-developed analysis of at least 3 supply-side policies that covers both types mentioned in preamble. Analysis is supported with the use and explanation of economic framework to explain outward shift of AS, before linking to improvement in BOT. Covers both price and non-price competitiveness Covers both TR_x and TE_M Use of relevant real-world examples in analysis.	8 - 10

Part (b)

Introduction - Clarify competitiveness

- Trade competitiveness - the ability of a country to sell goods and services competitively in a foreign country which is determined by both price and non-price factors such as product quality, design, reliability and performance, choice, after-sales services, marketing, branding and the availability and cost of replacement parts.
- Investment competitiveness - the ability to attract FDI which depends on the expected net rate of return (expected profitability) on investment projects, which is affected by macroeconomic and political stability of the country, size of the domestic market, availability of suitable labour, existence of good physical infrastructure and low tax rates.

Body

1. Briefly explain how a country may lose competitiveness

(a) Loss of trade competitiveness

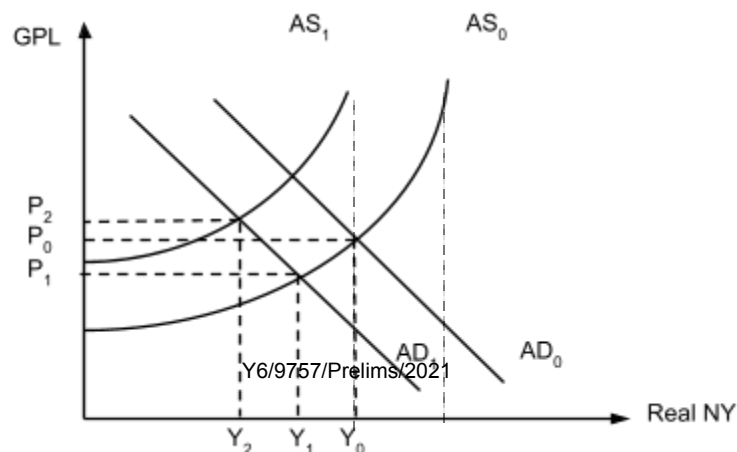
- Loss of CA due to depletion of natural resources, emergence of low-cost competitors, higher relative inflation rate (possibly due to higher unit labour costs when wages rise faster than labour productivity).
- Change in tastes and preferences.

(b) Loss of investment competitiveness

- Fall in labour productivity leading to rise in unit labour cost poor industrial relations leading to work stoppages and disruptions in work processes.
- Government policies, e.g. excessive burdensome rules and regulations for businesses, uncompetitive corporate tax rates, weak infrastructural support.
- Poor macroeconomic performance, e.g. weak EG rates, high inflation rates, depreciation of exchange rate.
- The above would generate poorer expected profit potential from investments in the country Deter inward FDI flows and may lead to more FDI outflows as well.

2. Explain challenges that arise from loss of competitiveness

- In the short-run, a fall in net X and I decreases autonomous AD from AD_0 to AD_1 as less production processes take place, leading to unplanned investment for firms which acts as a signal for them to step down production in the next period, as well as hiring less factors of production including labour causing unemployment to rise. The reduction in output leads to a fall in income and sets off the reverse multiplier process whereby the reduction of one man's spending reduces another man's income. The initial decrease in AD leads to many successive rounds of falls in induced consumption as households receiving lower incomes will spend less on domestically produced goods and services. However, the fall in each round of induced C diminishes in magnitude as some of the fall in household income is mitigated by the fall in leakages via taxes, savings and expenditure on imported goods and services. This process stops when the initial fall in autonomous AD is equal to the sum of the fall leakages. RNY decreases more than proportionately from Y_0 to Y_3 , representing a **fall in actual growth**.
- The loss in CA in exporting industries and the outflow of I can also bring about **structural unemployment**. The retrenched workers from the affected industries may not have the appropriate skills to be re-employed in other industries in the country Results in a mismatch between job opportunities and skills offered (occupational immobility) by the unemployed in the SR.
- The fall in net X revenue and net FDI outflow worsen both the current and capital and financial accounts respectively Worsens the country's overall **BOP position**.
- While a fall in investment contributes to slower capital formation in the economy, a "brain drain" slows the creation of innovative ideas, and thereafter the advancement of technology. This could create a secondary effect where investors might also lose confidence in the economy. There is a fall in both quantity and quality of resources of labour and capital.
- The fall in the quantity and quality of resources has the effect of limiting the growth in the country's productive capacity (i.e. the maximum amount of goods and services that the country can produce when all its resources are fully and efficiently utilized). This causes the AS curve to shift to the left from AS_0 to AS_1 with the full employment output level falling from Yf_0 to Yf_1 . Potential growth is lowered, making sustained growth more difficult to achieve, making the country more susceptible to inflation in the future.



0

 $Y_{f1} \quad Y_{f0}$

3. Factors that affect severity of challenges

- However, the extent to which the loss of competitiveness has a negative impact on the macroeconomic aims of a country depends on the following factors.

(a) Openness of economy: Relative size of AD components

- Countries with a larger domestic market may be able to mitigate the negative effects from the loss in competitiveness better. For instance, larger economies can engage in demand-management policies to prop up AD to mitigate the fall in RNY, as the size of domestic components of GDP is large relative to more open economies.
- In addition, they will also be affected less by any retaliatory measures from other countries for implementing any protectionistic measures.
- In contrast, small open economies cannot rely on C, I and G to prop up the fall in FDI and exports, as local firms with small domestic markets may not have enough profits to undertake significant investment to make up for the shortfall in FDI, and domestic consumption is unable to make up for the fall in exports.

(b) Effectiveness of policies to mitigate challenges

- If the government is able to effectively implement supply-side policies such as those mentioned in (a) to stem the loss of competitiveness, the negative impact on the economy will be alleviated. Supply-side policies such as education and retraining are also able to mitigate structural unemployment.
- However, there might be limitations to the supply-side policies that make their effectiveness differ across countries (*discuss any limitation by comparing two different scenarios to determine the success*).

(i) Uncertainty: The nature of supply-side policies to encourage R&D and retraining workers for new areas of CA is uncertain as it may not pay off. If governments do not pick the right winners, e.g. Malaysia with their support for the ailing car industry, compared to the Singapore government who has successfully transitioned the economy to high-end manufacturing such as pharmaceuticals and semiconductors, then the extent to which the negative impact of a loss of competitiveness will differ.

(ii) Govt budget: Governments facing a high level of debt (e.g. PIIGS in the EU) may be less able to implement tax cuts or spend on expansion of productive capacity, hence limiting the extent to which they can mitigate the loss of competitiveness and its negative impacts compared to governments with consistent budget surpluses.

(ii) Receptiveness of firms and workers: Most low-skilled workers tend to work long and irregular hours and do not have the time nor incentive to go for training while employers' attitudes towards training is usually lukewarm, thereby resulting in a low take-up rate. Also, if the new jobs available require skills that are too far-off from workers' current skill level, it will not be easy to upgrade or retrain them to be competent enough for the jobs, leading to difficulties in restoring competitiveness. However, if government is able to mitigate these by providing subsidies and grants for retraining, especially during a recession when labour is idle, the likelihood of success of retraining policies will be greater.

(c) State of economy

- If the economy is booming and is operating near full-capacity, the fall in AD goes towards relieving inflationary pressure as there is a reduction in the competition for factors of production, and there will be a smaller fall in RNY.
- Conversely, if the economy was operating on the horizontal range of the AS with a lot of slack, it will result in a larger fall in RNY and a larger rise in unemployment.

(d) Root cause

- If the root cause of the loss of competitiveness is due to an appreciation of exchange rate or a rise in GPL due to demand-pull inflation, the challenges will be less severe as it reflects the economy having strong fundamentals.
- On the other hand, if the loss of competitiveness is due to a loss of comparative advantage due to a persistent increase in wages in excess of productivity / depletion of natural resources / being leapfrogged in terms of price and quality by competitors, the resulting structural unemployment will compound the severity of SR challenges such as the fall in RNY and increase in cyclical UN.
- *Also acceptable: Depends on the persistence or duration of loss of competitiveness.*

Other acceptable factors:

(e) Size of multiplier

- Due to import dependence and mandatory savings in Singapore, MPM and MPS for Singapore is high compared to a relatively more self-sufficient economy such as the US. The higher marginal propensity to withdraw leads to a smaller k size.
- A fall in autonomous AD due to the loss of competitiveness brings about less successive rounds of falls in induced consumption, leading to a relatively smaller fall in RNY and rise in UN for Singapore compared to the US.

(f) Type of exports and size of industry

- Countries that export goods whose demand is price elastic (e.g. Singapore which exports high-end electronics which are non-necessities) will see a larger fall in quantity demanded given an increase in price, worsening BOT and the extent of fall in AD and hence RNY.
- On the contrary, countries that export goods that are price inelastic (e.g. agricultural goods), will face a smaller fall in quantity demanded of exports when prices rise, lessening the severity of the challenges faced such as EG.
- *Also acceptable: Explanations in the context of the size of industry that loses competitiveness.*

Conclusion and Overall Evaluation

- Overall, it is likely that the openness of economy is the predominant factor that determines the severity of potential challenges faced by a loss of competitiveness. In light of countries' increasing vulnerability to the emergence of low-cost competitors, larger countries like China are able to reduce their dependence on trade and FDI, and are focusing on boosting their domestic consumption and investment. These larger countries are also more able to implement demand-management policies as stopgap measures in order to stem the negative effects of a loss of competitiveness.
- However, in the case of Singapore, with a small domestic market, we have to focus on the effectiveness of supply-side policies to maintain our competitiveness instead of looking at demand-side measures. Being a small and open economy means that Singaporean firms and workers are often competing with the best in the world, and how far we can mitigate these challenges depends on how fast we restructure the economy against the rising pace of structural changes due to technological advancement and emergence of low-cost competitors.
- In the short-run the challenges may be more severe compared to the long-run as supply-side policies to allay these challenges have a long gestation period.

- Rising income inequality may also be a growing concern that leads to social instability as a result of the loss of competitiveness, worsening the negative impact. As such, policies that improve competitiveness should also be supplemented with social policies such as subsidies for education, healthcare and other means-tested benefits. This would not only ensure that our economy is competitive but also that growth in Singapore is inclusive.
- It is also noted that there are other factors that can affect trade flows that are unrelated to competitiveness such as proximity, preferential treatment from FTAs etc.

Mark Scheme:

Knowledge, Application, Understanding, Analysis		
L1	Answer attempts to present some arguments on the factors that affect the severity of challenges faced. Analysis is largely undeveloped and contains inaccuracies in content, or is one-sided without scope. No contextual knowledge to support arguments. Glaring conceptual errors.	1 – 4
L2	Answer presents well-balanced arguments on the factors that affect the severity of challenges faced, although the analysis may not be fully developed. Analysis is developed but may have gaps in arguments. Some attempt to contextualise using examples.	5 – 7
L3	Answer demonstrates a good understanding of the factors that affect the severity of challenges faced. Economic analysis is well-developed throughout with and argument is coherent. At least two factors well-discussed. Answer makes use of real-life examples to support arguments.	8 – 10
Evaluation		
E1	An unexplained judgement An unexplained evaluative conclusion/comment	1
E2	Evaluative assessment supported by economic analysis Substantiation of an evaluative comment and/or conclusion	3 – 4
E3	Able to give in depth or well explained insights, e.g. elaboration by weighing which factor is most likely to be the predominant one.	4– 5

*******END*******